Report on Glacier Data Prediction

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1. INTRODUCTION

This report presents the data processing and modeling workflow applied to the glacier grid dataset, which includes:

- Raw data ingestion and cleaning
- Time-based split: training set / test set
- EDA
- DATA PREPROCESSING
- Model training and comparison: Linear Regression, Lasso, SVR, Decision Tree, Random Forest, MLP
- Using Genetic Algorithm (GA) to Optimize Feature Selection

The code and results are available on GitHub (https://github.com/ten-z/Evolutionary-Machine-Learning). Please note that the High-res-more-variables dataset was too large to process locally within the available time, so it has not been included in this report.

2. RAW DATA INGESTION AND CLEANING

2.1. Sequential Reading and Initial Cleaning

Read each year's raw gridded data in chronological order.

Remove any rows containing the fill value (9.969…e+36) or NaN, which correspond almost entirely to border cells over open ocean.

2.2. Local Neighborhood Feature Construction

Manually specify one of ice_thickness, ice_velocity, or ice_mask as the prediction target ("label").

For each remaining grid cell, collect all non-coordinate variables from its eight surrounding neighbors.

If a neighbor has been removed (open ocean), substitute zeros for its features.

Append these neighborhood values as new features to the central cell.

2.3. Historical Feature Augmentation

For each grid cell, retrieve the same set of variables at the identical location over the past K years (default past_years = 3).

If a previous year's data is unavailable, fill those historic features with zeros.

Append these K-year historical values as additional features.

2.4. Year Feature and Final Assembly

Combine the processed data from all years into a single table.

Insert a new year column at the front of every row to record its original year.

Export the consolidated dataset to {label}_all_years.csv.

3. TIME-BASED SPLIT: TRAINING SET / TEST SET

Sort the data in ascending order by year, then split it into a training set and a test set according to the TRAIN_RATIO (default 0.8), ensuring that all training-set years come before any test-set years and that at least one year is reserved for testing. Finally, save the training and test sets to separate CSV files.

4. EDA

4.1. Basic Statistics

In datasplitandEDA.py, output the following logs for the raw training data: data.head(), data.info(), data.describe(), number of missing values, and proportion of missing values. Save the resulting file as data_describe_summary.csv in each dataset's directory.

Below are the results for the Low-res-less-variables and Low-res-more-variables datasets using ice_thickness and ice_velocity as the labels, respectively:

data_describe_summary

	count	mean	std	min	25%	50%	75%	max
year	62605.0	2048.356680776300	19.62121062931100	2015.0	2031.0	2048.0	2065.0	2082.0
x_coordinate	62605.0	451783.6722306530	1172030.3381903600	-2432000.0	-486400.0	608000.0	1459200.0	2675200.0
y_coordinate	62605.0	77489.52959028830	1003126.9153748500	-2067200.0	-729600.0	121600.0	851200.0	2188800.0
precipitation	62605.0	208.8470233918	276.77546053600900	-108.359268188	44.2476844788	98.0099639893	251.934860229	2459.45605469
air_temperature	62605.0	237.54867608379800	11.510238339904000	214.175415039	227.551956177	237.588165283	246.036178589	273.111785889
ocean_temperature	62605.0	272.325862614285	0.5461632284764640	271.191192627	271.924499512	272.236175537	272.607666016	275.000488281
upper_left_precipitation	62605.0	164.81362225985300	231.6699438772850	-108.359268188	35.0633239746	78.2778091431	196.929824829	2459.45605469
upper_left_air_temperature	62605.0	219.36646871074000	61.936116205963500	0.0	225.278747559	234.288879395	244.305114746	273.111785889
upper_left_ocean_temperature	62605.0	252.7416289978540	70.32136837272450	0.0	271.849975586	272.169891357	272.550689697	274.443939209
top_precipitation	62605.0	169.29598435156000	225.40547570005300	-108.359268188	36.7137947083	81.8392944336	208.366394043	2294.9050293
top_air_temperature	62605.0	222.93057239496800	56.12383619100130	0.0	225.724060059	234.980285645	244.729644775	266.438110352
top_ocean_temperature	62605.0	256.64304964632100	63.44172866501870	0.0	271.86807251	272.191040039	272.580963135	275.000488281
upper_right_precipitation	62605.0	156.8769297216300	203.94471057527100	-108.359268188	34.6502075195	77.7619018555	196.60144043	1816.53466797
upper_right_air_temperature	62605.0	218.1699912408870	63.701822615928400	0.0	225.122589111	234.143707275	244.340072632	266.81262207
upper_right_ocean_temperature	62605.0	251.4692674197980	72.44206897295090	0.0	271.859710693	272.193359375	272.585449219	275.000488281
left_precipitation	62605.0	180.04837362999000	258.2912787775290	-108.359268188	37.4941062927	83.0801620483	209.141571045	2459.45605469
left_air_temperature	62605.0	224.49185184710000	53.2228276918421	0.0	225.923843384	235.270935059	244.717330933	273.111785889
left_ocean_temperature	62605.0	258.3669285176100	60.00750233374530	0.0	271.87008667	272.185577393	272.558532715	274.274291992
right_precipitation	62605.0	170.78114176306000	218.0682936711750	-108.359268188	37.5914649963	83.5234375	212.064819336	1967.79467773
right_air_temperature	62605.0	224.53498979648700	53.23638519953510	0.0	225.923843384	235.271621704	244.829086304	267.590484619
right_ocean_temperature	62605.0	258.39510527197200	60.014204578905900	0.0	271.886352539	272.209533691	272.592834473	275.000488281
lower_left_precipitation	62605.0	171.89783408031800	255.56585453653500	-108.359268188	34.5181045532	76.7882995605	191.321578979	2342.63427734
lower_left_air_temperature	62605.0	218.12434242162600	63.69672178732290	0.0	225.122589111	234.052459717	244.096954346	273.111785889
lower_left_ocean_temperature	62605.0	251.42454401946500	72.42888222266600	0.0	271.849884033	272.160675049	272.522857666	274.274291992
bottom_precipitation	62605.0	180.94391006889800	261.6973300900000	-65.8212127686	36.7096328735	81.4745483398	203.799255371	2459.45605469
bottom_air_temperature	62605.0	222.9005712210120	56.12077461682580	0.0	225.724060059	234.980285645	244.536315918	273.111785889
bottom_ocean_temperature	62605.0	256.62874721493300	63.437981616239500	0.0	271.873260498	272.187225342	272.559509277	275.000488281
lower_right_precipitation	62605.0	167.83317732762500	231.61369374538600	-108.359268188	35.0568275452	77.9970092773	196.071472168	2335.80810547
lower_right_air_temperature	62605.0	219.33260947292000	61.918369125347600	0.0	225.278747559	234.288879395	244.258377075	267.590484619
lower_right_ocean_temperature	62605.0	252.75712894086100	70.32567005052600	0.0	271.868286133	272.18850708	272.569244385	275.000488281
precipitation_3_years_ago	62605.0	197.78007804839300	272.54389754908800	-108.359268188	39.0327529907	89.2989196777	237.613952637	2459.45605469
air_temperature_3_years_ago	62605.0	226.54826353664300	50.98969743425210	0.0	226.329376221	236.395858765	245.630111694	273.111785889
ocean_temperature_3_years_ago	62605.0	259.7934481550490	57.04131552857310	0.0	271.876647949	272.195770264	272.577667236	275.000488281
precipitation_2_years_ago	62605.0	201.40497763691400	273.7873138900730	-108.359268188	40.6939849854	92.4044342041	242.837615967	2459.45605469
air_temperature_2_years_ago	62605.0	230.17174722710900	42.582963316095900	0.0	226.73008728	236.768692017	245.772155762	273.111785889
ocean_temperature_2_years_ago	62605.0	263.9239060814360	47.07431894759470	0.0	271.892456055	272.209106445	272.58706665	275.000488281
precipitation_1_year_ago	62605.0	205.1605672779940	275.3511394241540	-108.359268188	42.5544586182	95.2460098267	247.490127563	2459.45605469
air_temperature_1_year_ago	62605.0	233.81507567360000	31.57444256199730	0.0	227.168258667	237.182388306	245.913543701	273.111785889
ocean_temperature_1_year_ago	62605.0	268.07264596405800	33.75746988720320	0.0	271.90814209	272.223449707	272.597076416	275.000488281
ice_thickness	62605.0	1918.8392251958700	1071.6295773098800	0.0100353881717	982.284790039	2073.45605469	2837.94873047	4586.17773438

4.1.1 Low-res-less-variables dataset with ice_thickness as the label.

data_describe_summary

	count	mean	std	min	25%	50%	75%	max
year	62605.0	2048.356680776300	19.62121062931100	2015.0	2031.0	2048.0	2065.0	2082.0
x_coordinate	62605.0	451783.6722306530	1172030.3381903600	-2432000.0	-486400.0	608000.0	1459200.0	2675200.0
y_coordinate	62605.0	77489.52959028830	1003126.9153748500	-2067200.0	-729600.0	121600.0	851200.0	2188800.0
precipitation	62605.0	208.8470233918	276.77546053600900	-108.359268188	44.2476844788	98.0099639893	251.934860229	2459.45605469
air_temperature	62605.0	237.54867608379800	11.510238339904000	214.175415039	227.551956177	237.588165283	246.036178589	273.111785889
ocean_temperature	62605.0	272.325862614285	0.5461632284764640	271.191192627	271.924499512	272.236175537	272.607666016	275.000488281
upper_left_precipitation	62605.0	164.81362225985300	231.6699438772850	-108.359268188	35.0633239746	78.2778091431	196.929824829	2459.45605469
upper_left_air_temperature	62605.0	219.36646871074000	61.936116205963500	0.0	225.278747559	234.288879395	244.305114746	273.111785889
upper_left_ocean_temperature	62605.0	252.7416289978540	70.32136837272450	0.0	271.849975586	272.169891357	272.550689697	274.443939209
top_precipitation	62605.0	169.29598435156000	225.40547570005300	-108.359268188	36.7137947083	81.8392944336	208.366394043	2294.9050293
top_air_temperature	62605.0	222.93057239496800	56.12383619100130	0.0	225.724060059	234.980285645	244.729644775	266.438110352
top_ocean_temperature	62605.0	256.64304964632100	63.44172866501870	0.0	271.86807251	272.191040039	272.580963135	275.000488281
upper_right_precipitation	62605.0	156.8769297216300	203.94471057527100	-108.359268188	34.6502075195	77.7619018555	196.60144043	1816.53466797
upper_right_air_temperature	62605.0	218.1699912408870	63.701822615928400	0.0	225.122589111	234.143707275	244.340072632	266.81262207
upper_right_ocean_temperature	62605.0	251.4692674197980	72.44206897295090	0.0	271.859710693	272.193359375	272.585449219	275.000488281
left_precipitation	62605.0	180.04837362999000	258.2912787775290	-108.359268188	37.4941062927	83.0801620483	209.141571045	2459.45605469
left_air_temperature	62605.0	224.49185184710000	53.2228276918421	0.0	225.923843384	235.270935059	244.717330933	273.111785889
left_ocean_temperature	62605.0	258.3669285176100	60.00750233374530	0.0	271.87008667	272.185577393	272.558532715	274.274291992
right_precipitation	62605.0	170.78114176306000	218.0682936711750	-108.359268188	37.5914649963	83.5234375	212.064819336	1967.79467773
right_air_temperature	62605.0	224.53498979648700	53.23638519953510	0.0	225.923843384	235.271621704	244.829086304	267.590484619
right_ocean_temperature	62605.0	258.39510527197200	60.014204578905900	0.0	271.886352539	272.209533691	272.592834473	275.000488281
lower_left_precipitation	62605.0	171.89783408031800	255.56585453653500	-108.359268188	34.5181045532	76.7882995605	191.321578979	2342.63427734
lower_left_air_temperature	62605.0	218.12434242162600	63.69672178732290	0.0	225.122589111	234.052459717	244.096954346	273.111785889
lower_left_ocean_temperature	62605.0	251.42454401946500	72.42888222266600	0.0	271.849884033	272.160675049	272.522857666	274.274291992
bottom_precipitation	62605.0	180.94391006889800	261.6973300900000	-65.8212127686	36.7096328735	81.4745483398	203.799255371	2459.45605469
bottom_air_temperature	62605.0	222.9005712210120	56.12077461682580	0.0	225.724060059	234.980285645	244.536315918	273.111785889
bottom_ocean_temperature	62605.0	256.62874721493300	63.437981616239500	0.0	271.873260498	272.187225342	272.559509277	275.000488281
lower_right_precipitation	62605.0	167.83317732762500	231.61369374538600	-108.359268188	35.0568275452	77.9970092773	196.071472168	2335.80810547
lower_right_air_temperature	62605.0	219.33260947292000	61.918369125347600	0.0	225.278747559	234.288879395	244.258377075	267.590484619
lower_right_ocean_temperature	62605.0	252.75712894086100	70.32567005052600	0.0	271.868286133	272.18850708	272.569244385	275.000488281
precipitation_3_years_ago	62605.0	197.78007804839300	272.54389754908800	-108.359268188	39.0327529907	89.2989196777	237.613952637	2459.45605469
air_temperature_3_years_ago	62605.0	226.54826353664300	50.98969743425210	0.0	226.329376221	236.395858765	245.630111694	273.111785889
ocean_temperature_3_years_ago	62605.0	259.7934481550490	57.04131552857310	0.0	271.876647949	272.195770264	272.577667236	275.000488281
precipitation_2_years_ago	62605.0	201.40497763691400	273.7873138900730	-108.359268188	40.6939849854	92.4044342041	242.837615967	2459.45605469
air_temperature_2_years_ago	62605.0	230.17174722710900	42.582963316095900	0.0	226.73008728	236.768692017	245.772155762	273.111785889
ocean_temperature_2_years_ago	62605.0	263.9239060814360	47.07431894759470	0.0	271.892456055	272.209106445	272.58706665	275.000488281
precipitation_1_year_ago	62605.0	205.1605672779940	275.3511394241540	-108.359268188	42.5544586182	95.2460098267	247.490127563	2459.45605469
air_temperature_1_year_ago	62605.0	233.81507567360000	31.57444256199730	0.0	227.168258667	237.182388306	245.913543701	273.111785889
ocean_temperature_1_year_ago	62605.0	268.07264596405800	33.75746988720320	0.0	271.90814209	272.223449707	272.597076416	275.000488281
ice_velocity	62605.0	86.09490362408030	308.1544644910150	0.0	2.61102294922	8.17558765411	27.8806819916	13779.2851562

4.1.2 Low-res-less-variables dataset with ice_velocity as the label.

data_describe_summary

	count	mean	std	min	25%	50%	75%	max
year	62331.0	2048.1888626847000	19.585793738218900	2015.0	2031.0	2048.0	2065.0	2082.0
x_coordinate	62331.0	478360.4434390590	1156753.9341951100	-2432000.0	-364800.0	608000.0	1459200.0	2675200.0
y_coordinate	62331.0	51647.46915659940	1009954.8271016900	-2067200.0	-729600.0	0.0	851200.0	2067200.0
bed_rock_elevation	62331.0	6.943450841143880	716.2299487900100	-2251.47460938	-457.963058472	-16.4050750732	438.632659912	2637.58935547
precipitation	62331.0	229.9422943186040	285.699102556493	-2008.7409668	53.38990592955	116.700912476	283.9609680175	2366.13134766
air_temperature	62331.0	238.76057524120300	11.881583639308400	214.40524292	228.602180481	238.788574219	247.77316284150000	277.132171631
ocean_temperature	62331.0	272.51512881407800	0.6448134691218480	271.108581543	272.0337677	272.380462646	272.865234375	277.42578125
upper_left_bed_rock_elevation	62331.0	25.459247840682100	702.2440796341410	-2251.47460938	-422.45655822750000	0.0	434.196533203	2637.58935547
upper_left_precipitation	62331.0	182.2244567150450	240.46653924125300	-2008.7409668	43.293296814	92.4893264771	222.9855575565	2366.13134766
upper_left_air_temperature	62331.0	220.2992069819480	62.43301097808270	0.0	226.1079483035	235.387329102	245.717536926	277.132171631
upper_left_ocean_temperature	62331.0	252.81820529381600	70.56903777541590	0.0	271.9682159425	272.318939209	272.8181152345	277.42578125
top_bed_rock_elevation	62331.0	10.837095542608300	699.2069482820320	-2251.47460938	-438.779571533	0.0	430.4826812745	2637.58935547
top_precipitation	62331.0	187.57982384431500	235.21725122416400	-1062.90881348	45.18831062315	97.1563873291	233.70187377900000	1840.85339355
top_air_temperature	62331.0	224.06550860009700	56.32663413735040	0.0	226.657814026	236.02633667	246.19286346450000	269.322906494
top_ocean_temperature	62331.0	256.89986071018000	63.353631768814200	0.0	271.987609863	272.336090088	272.836746216	276.630218506
upper_right_bed_rock_elevation	62331.0	10.061623876486700	689.2015702334290	-2251.47460938	-432.699768066	0.0	421.713806152	2637.58935547
upper_right_precipitation	62331.0	175.75801451662100	219.86071866660400	-963.238830566	42.62647247315	91.3712768555	222.266395569	1840.85339355
upper_right_air_temperature	62331.0	218.84207945648200	64.53154905327360	0.0	225.9237976075	235.1743927	245.7166366575	268.714263916
upper_right_ocean_temperature	62331.0	251.27492029022900	73.08195754016000	0.0	271.972869873	272.327911377	272.82904052750000	276.630218506
left_bed_rock_elevation	62331.0	12.998223072090600	708.5365107917870	-2251.47460938	-445.932952881	0.0	434.196533203	2637.58935547
left_precipitation	62331.0	199.49239787858500	267.62271512401	-2008.7409668	45.86624717715	98.8014602661	238.103065491	2366.13134766
left_air_temperature	62331.0	225.25277315975600	54.223437256288800	0.0	226.818702698	236.236862183	246.21331024200000	277.132171631
left_ocean_temperature	62331.0	258.1589107297720	60.84654136510820	0.0	271.982513428	272.324615479	272.80732727050000	277.42578125
right_bed_rock_elevation	62331.0	9.08207093824249	698.5991103357730	-2251.47460938	-439.624481201	0.0	432.115661621	2637.58935547
right_precipitation	62331.0	191.28613525682100	235.17932276157600	-963.238830566	45.8964290619	99.1968917847	239.6359024045	2084.96850586
right_air_temperature	62331.0	225.28137408772900	54.22850570439950	0.0	226.818702698	236.240325928	246.3420333865	268.826538086
right_ocean_temperature	62331.0	258.1756733704060	60.85055057821640	0.0	271.990600586	272.339935303	272.833404541	276.630218506
lower_left_bed_rock_elevation	62331.0	2.273100633394420	697.8420177683600	-2251.47460938	-445.932952881	0.0	407.416503906	2637.58935547
lower_left_precipitation	62331.0	189.66777945570600	261.34611602696500	-1167.89672852	42.57806396485	90.9635620117	220.857444763	2307.68530273
lower_left_air_temperature	62331.0	218.8840694894750	64.56811315359150	0.0	225.9237976075	235.144989014	245.5677108765	276.782897949
lower_left_ocean_temperature	62331.0	251.21620030778300	73.06455080019240	0.0	271.946289062	272.280975342	272.750396729	276.737030029
bottom_bed_rock_elevation	62331.0	11.02738263262280	703.6532369362760	-2251.47460938	-447.032989502	0.0	432.447662354	2637.58935547
bottom_precipitation	62331.0	200.4740332277730	270.81476490581500	-2008.7409668	45.27878189085	97.2990341187	234.6404037475	2366.13134766
bottom_air_temperature	62331.0	224.11815122823500	56.365737103504800	0.0	226.657814026	236.02633667	246.09169769300000	277.132171631
bottom_ocean_temperature	62331.0	256.86522432047200	63.34480596720430	0.0	271.97505188	272.312133789	272.789611816	277.42578125
lower_right_bed_rock_elevation	62331.0	8.18227141465005	692.3908978331870	-2251.47460938	-439.714050293	0.0	427.168823242	2637.58935547
lower_right_precipitation	62331.0	187.23383288308300	243.23505034174400	-963.238830566	43.291122436500000	92.6796569824	226.509788513	2084.96850586
lower_right_air_temperature	62331.0	220.35055586693100	62.457418019742600	0.0	226.1079483035	235.391479492	245.817062378	270.591827393
lower_right_ocean_temperature	62331.0	252.8036020385470	70.56480417524780	0.0	271.96635437	272.308563232	272.796691895	276.630218506
bed_rock_elevation_3_years_ago	62331.0	7.822553785821650	699.5790000520230	-2251.47460938	-439.762115479	0.0	427.729644775	2637.58935547
precipitation_3_years_ago	62331.0	217.9139913843010	282.2576562548440	-1879.72705078	47.5251045227	106.965057373	267.2853393555	2366.13134766
air_temperature_3_years_ago	62331.0	227.5395743324810	51.4523770077457	0.0	227.236534119	237.395294189	247.2003250125	276.966064453
ocean_temperature_3_years_ago	62331.0	259.8789061432690	57.26510723233820	0.0	271.983612061	272.327484131	272.810623169	277.307037354
bed_rock_elevation_2_years_ago	62331.0	7.6215659884115000	705.1097002598140	-2251.47460938	-447.65222168	0.0	432.447662354	2637.58935547
precipitation_2_years_ago	62331.0	222.12650545995400	283.61344651475200	-1879.72705078	49.6122970581	110.147209167	273.3998260495	2366.13134766
air_temperature_2_years_ago	62331.0	231.23243289087000	43.00225347293380	0.0	227.6757049565	237.855422974	247.3953475955	276.966064453
ocean_temperature_2_years_ago	62331.0	264.0422633553110	47.266261000058500	0.0	272.000640869	272.344726562	272.829421997	277.307037354
bed_rock_elevation_1_year_ago	62331.0	7.362801642834000	710.6547231089960	-2251.47460938	-451.53758239750000	-1.4634540081	436.17376709	2637.58935547
precipitation_1_year_ago	62331.0	225.81700356429500	284.56859859723500	-2008.7409668	51.51787567135	113.364051819	278.18309021	2366.13134766
air_temperature_1_year_ago	62331.0	234.9387543509930	31.96945427254820	0.0	228.130187988	238.318695068	247.591644287	277.132171631
ocean_temperature_1_year_ago	62331.0	268.21520862382900	33.94023786953200	0.0	272.017333984	272.362060547	272.847351074	277.42578125
ice_thickness	62331.0	1917.8634570688900	1077.8319296910100	0.0104743046686	967.819152832	2080.35620117	2843.658569335	4586.45166016

4.1.3 Low-res-more-variables dataset with ice_thickness as the label.

data_describe_summary

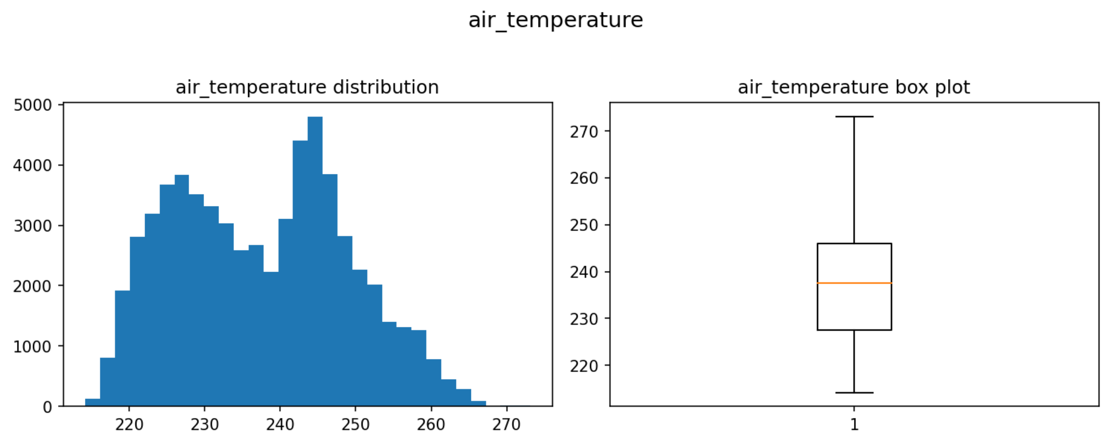
	count	mean	std	min	25%	50%	75%	max
year	62331.0	2048.1888626847000	19.585793738218900	2015.0	2031.0	2048.0	2065.0	2082.0
x_coordinate	62331.0	478360.4434390590	1156753.9341951100	-2432000.0	-364800.0	608000.0	1459200.0	2675200.0
y_coordinate	62331.0	51647.46915659940	1009954.8271016900	-2067200.0	-729600.0	0.0	851200.0	2067200.0
bed_rock_elevation	62331.0	6.943450841143880	716.2299487900100	-2251.47460938	-457.963058472	-16.4050750732	438.632659912	2637.58935547
precipitation	62331.0	229.9422943186040	285.699102556493	-2008.7409668	53.38990592955	116.700912476	283.9609680175	2366.13134766
air_temperature	62331.0	238.76057524120300	11.881583639308400	214.40524292	228.602180481	238.788574219	247.77316284150000	277.132171631
ocean_temperature	62331.0	272.51512881407800	0.6448134691218480	271.108581543	272.0337677	272.380462646	272.865234375	277.42578125
upper_left_bed_rock_elevation	62331.0	25.459247840682100	702.2440796341410	-2251.47460938	-422.45655822750000	0.0	434.196533203	2637.58935547
upper_left_precipitation	62331.0	182.2244567150450	240.46653924125300	-2008.7409668	43.293296814	92.4893264771	222.9855575565	2366.13134766
upper_left_air_temperature	62331.0	220.2992069819480	62.43301097808270	0.0	226.1079483035	235.387329102	245.717536926	277.132171631
upper_left_ocean_temperature	62331.0	252.81820529381600	70.56903777541590	0.0	271.9682159425	272.318939209	272.8181152345	277.42578125
top_bed_rock_elevation	62331.0	10.837095542608300	699.2069482820320	-2251.47460938	-438.779571533	0.0	430.4826812745	2637.58935547
top_precipitation	62331.0	187.57982384431500	235.21725122416400	-1062.90881348	45.18831062315	97.1563873291	233.70187377900000	1840.85339355
top_air_temperature	62331.0	224.06550860009700	56.32663413735040	0.0	226.657814026	236.02633667	246.19286346450000	269.322906494
top_ocean_temperature	62331.0	256.89986071018000	63.353631768814200	0.0	271.987609863	272.336090088	272.836746216	276.630218506
upper_right_bed_rock_elevation	62331.0	10.061623876486700	689.2015702334290	-2251.47460938	-432.699768066	0.0	421.713806152	2637.58935547
upper_right_precipitation	62331.0	175.75801451662100	219.86071866660400	-963.238830566	42.62647247315	91.3712768555	222.266395569	1840.85339355
upper_right_air_temperature	62331.0	218.84207945648200	64.53154905327360	0.0	225.9237976075	235.1743927	245.7166366575	268.714263916
upper_right_ocean_temperature	62331.0	251.27492029022900	73.08195754016000	0.0	271.972869873	272.327911377	272.82904052750000	276.630218506
left_bed_rock_elevation	62331.0	12.998223072090600	708.5365107917870	-2251.47460938	-445.932952881	0.0	434.196533203	2637.58935547
left_precipitation	62331.0	199.49239787858500	267.62271512401	-2008.7409668	45.86624717715	98.8014602661	238.103065491	2366.13134766
left_air_temperature	62331.0	225.25277315975600	54.223437256288800	0.0	226.818702698	236.236862183	246.21331024200000	277.132171631
left_ocean_temperature	62331.0	258.1589107297720	60.84654136510820	0.0	271.982513428	272.324615479	272.80732727050000	277.42578125
right_bed_rock_elevation	62331.0	9.08207093824249	698.5991103357730	-2251.47460938	-439.624481201	0.0	432.115661621	2637.58935547
right_precipitation	62331.0	191.28613525682100	235.17932276157600	-963.238830566	45.8964290619	99.1968917847	239.6359024045	2084.96850586
right_air_temperature	62331.0	225.28137408772900	54.22850570439950	0.0	226.818702698	236.240325928	246.3420333865	268.826538086
right_ocean_temperature	62331.0	258.1756733704060	60.85055057821640	0.0	271.990600586	272.339935303	272.833404541	276.630218506
lower_left_bed_rock_elevation	62331.0	2.273100633394420	697.8420177683600	-2251.47460938	-445.932952881	0.0	407.416503906	2637.58935547
lower_left_precipitation	62331.0	189.66777945570600	261.34611602696500	-1167.89672852	42.57806396485	90.9635620117	220.857444763	2307.68530273
lower_left_air_temperature	62331.0	218.8840694894750	64.56811315359150	0.0	225.9237976075	235.144989014	245.5677108765	276.782897949
lower_left_ocean_temperature	62331.0	251.21620030778300	73.06455080019240	0.0	271.946289062	272.280975342	272.750396729	276.737030029
bottom_bed_rock_elevation	62331.0	11.02738263262280	703.6532369362760	-2251.47460938	-447.032989502	0.0	432.447662354	2637.58935547
bottom_precipitation	62331.0	200.4740332277730	270.81476490581500	-2008.7409668	45.27878189085	97.2990341187	234.6404037475	2366.13134766
bottom_air_temperature	62331.0	224.11815122823500	56.365737103504800	0.0	226.657814026	236.02633667	246.09169769300000	277.132171631
bottom_ocean_temperature	62331.0	256.86522432047200	63.34480596720430	0.0	271.97505188	272.312133789	272.789611816	277.42578125
lower_right_bed_rock_elevation	62331.0	8.18227141465005	692.3908978331870	-2251.47460938	-439.714050293	0.0	427.168823242	2637.58935547
lower_right_precipitation	62331.0	187.23383288308300	243.23505034174400	-963.238830566	43.291122436500000	92.6796569824	226.509788513	2084.96850586
lower_right_air_temperature	62331.0	220.35055586693100	62.457418019742600	0.0	226.1079483035	235.391479492	245.817062378	270.591827393
lower_right_ocean_temperature	62331.0	252.8036020385470	70.56480417524780	0.0	271.96635437	272.308563232	272.796691895	276.630218506
bed_rock_elevation_3_years_ago	62331.0	7.822553785821650	699.5790000520230	-2251.47460938	-439.762115479	0.0	427.729644775	2637.58935547
precipitation_3_years_ago	62331.0	217.9139913843010	282.2576562548440	-1879.72705078	47.5251045227	106.965057373	267.2853393555	2366.13134766
air_temperature_3_years_ago	62331.0	227.5395743324810	51.4523770077457	0.0	227.236534119	237.395294189	247.2003250125	276.966064453
ocean_temperature_3_years_ago	62331.0	259.8789061432690	57.26510723233820	0.0	271.983612061	272.327484131	272.810623169	277.307037354
bed_rock_elevation_2_years_ago	62331.0	7.6215659884115000	705.1097002598140	-2251.47460938	-447.65222168	0.0	432.447662354	2637.58935547
precipitation_2_years_ago	62331.0	222.12650545995400	283.61344651475200	-1879.72705078	49.6122970581	110.147209167	273.3998260495	2366.13134766
air_temperature_2_years_ago	62331.0	231.23243289087000	43.00225347293380	0.0	227.6757049565	237.855422974	247.3953475955	276.966064453
ocean_temperature_2_years_ago	62331.0	264.0422633553110	47.266261000058500	0.0	272.000640869	272.344726562	272.829421997	277.307037354
bed_rock_elevation_1_year_ago	62331.0	7.362801642834000	710.6547231089960	-2251.47460938	-451.53758239750000	-1.4634540081	436.17376709	2637.58935547
precipitation_1_year_ago	62331.0	225.81700356429500	284.56859859723500	-2008.7409668	51.51787567135	113.364051819	278.18309021	2366.13134766
air_temperature_1_year_ago	62331.0	234.9387543509930	31.96945427254820	0.0	228.130187988	238.318695068	247.591644287	277.132171631
ocean_temperature_1_year_ago	62331.0	268.21520862382900	33.94023786953200	0.0	272.017333984	272.362060547	272.847351074	277.42578125
ice_velocity	62331.0	92.60670759125770	367.6631323825370	0.0	2.62156379223	8.29279899597	28.34227180485	19250.7460938

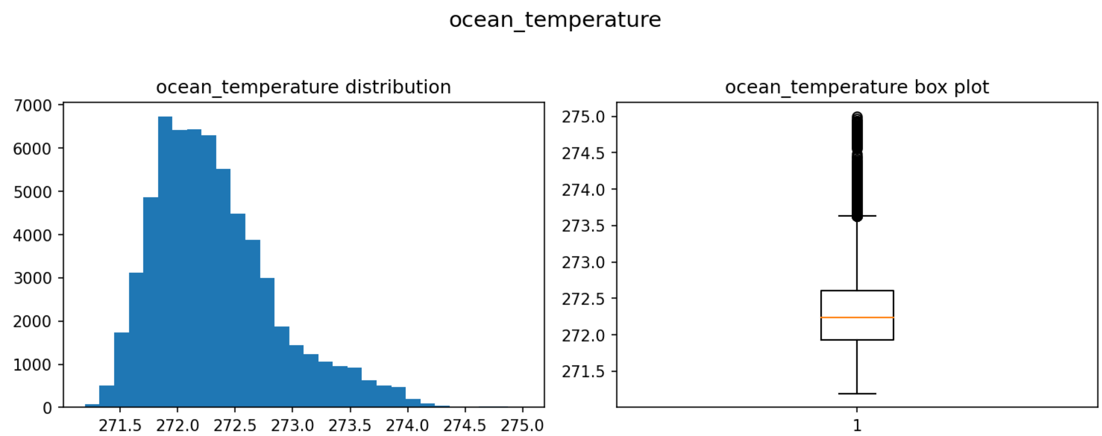
4.1.4 Low-res-more-variables dataset with ice_velocity as the label.

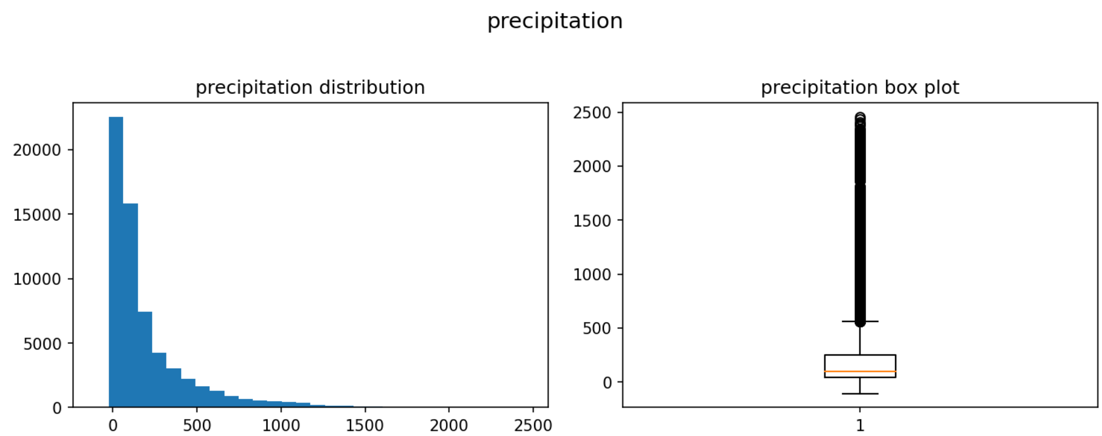
4.2. Variable Distribution

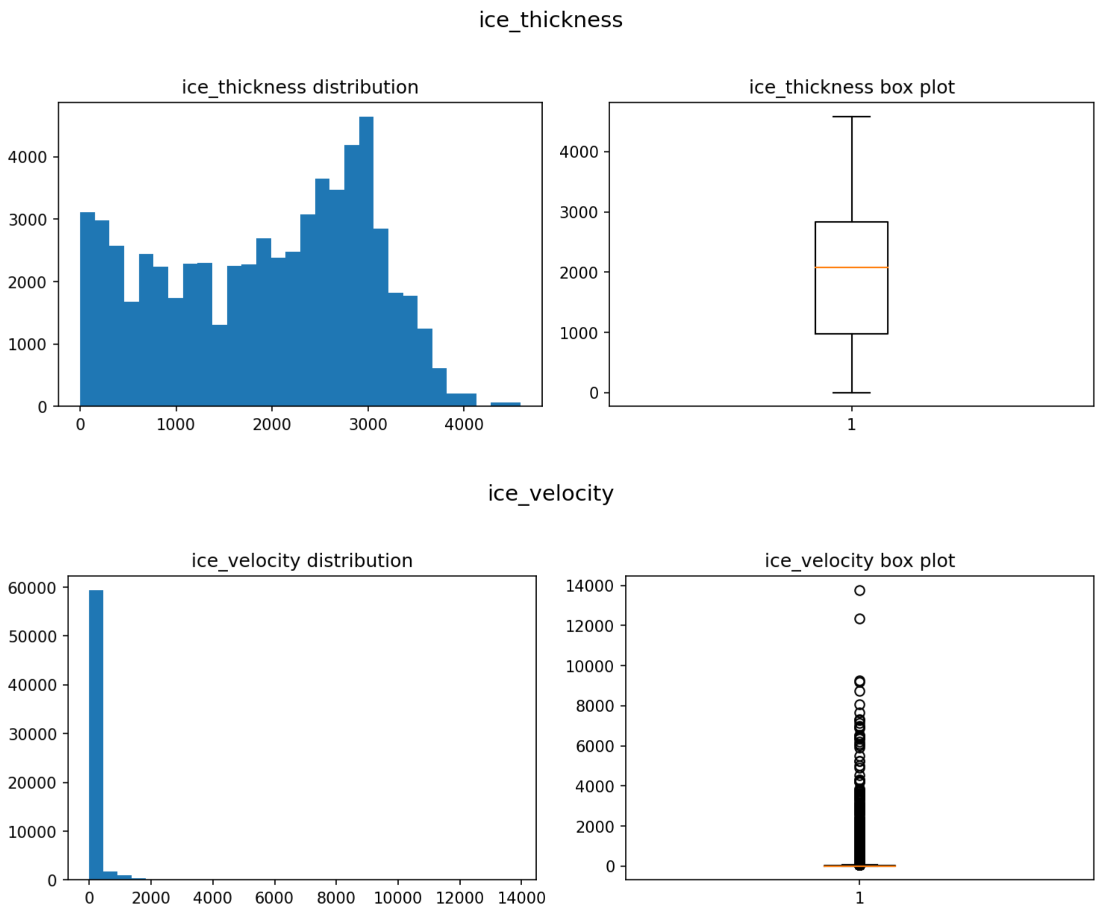
The distribution plot for each variable in the training set is stored in the “distribution_plot” directory of that dataset.

Below are the distribution plots of several variables from the Low-res-less-variables dataset and Low-res-more-variables dataset:

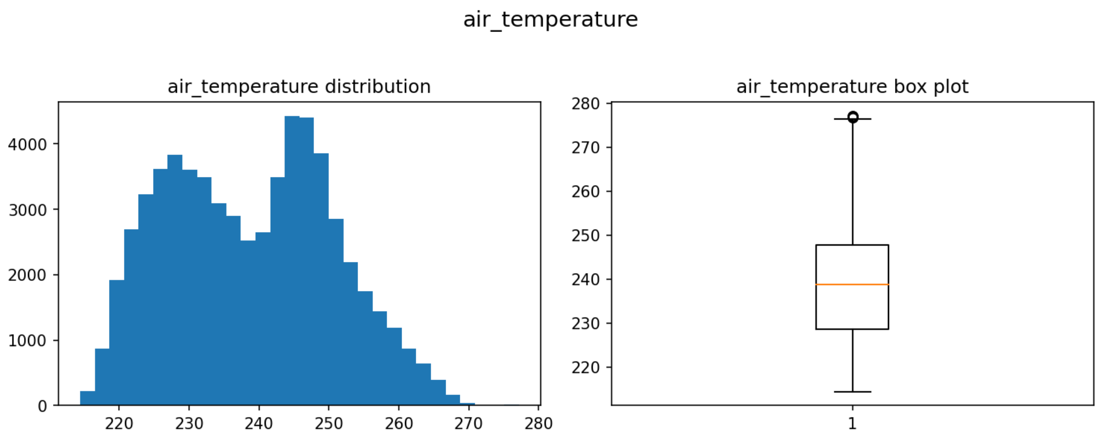

air_temperature
air_temperature distribution
5000
4000
3000
2000
1000
0
220 230 240 250 260 270
air_temperature box plot
270
260
250
240
230
220
1


ocean_temperature
ocean_temperature distribution
7000
6000
5000
4000
3000
2000
1000
0
271.5 272.0 272.5 273.0 273.5 274.0 274.5 275.0
ocean_temperature box plot
275.0
274.5
274.0
273.5
273.0
272.5
272.0
271.5
1

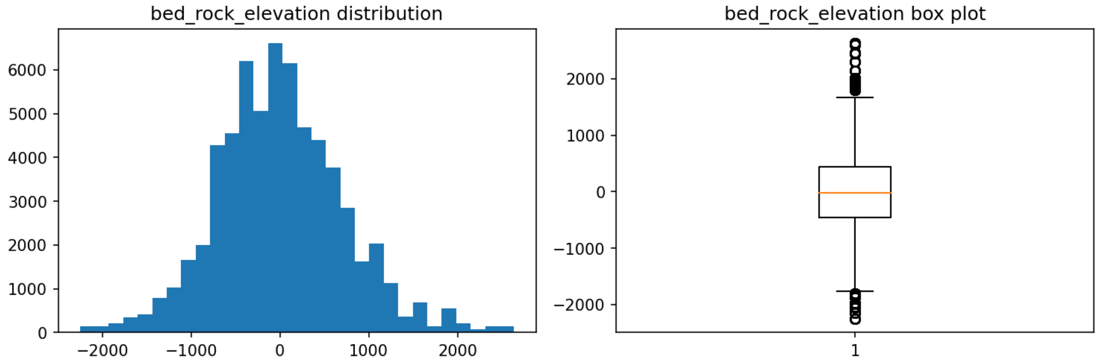

precipitation
precipitation distribution
20000
15000
10000
5000
0
0 500 1000 1500 2000 2500
precipitation box plot
2500
2000
1500
1000
500
0
1



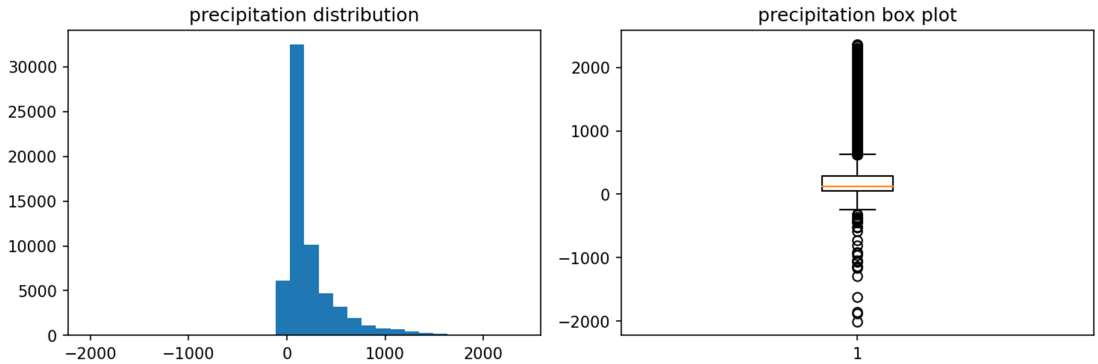
4.2.1 distribution plots from the Low-res-less-variables dataset



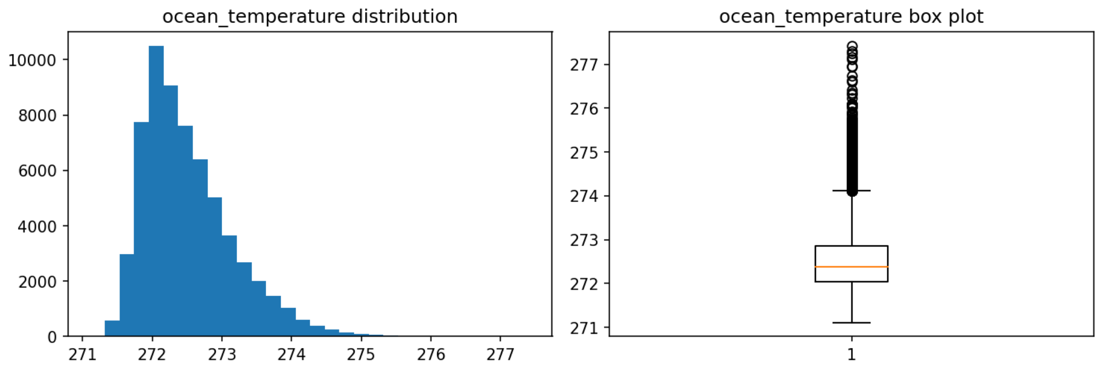
bed_rock_elevation

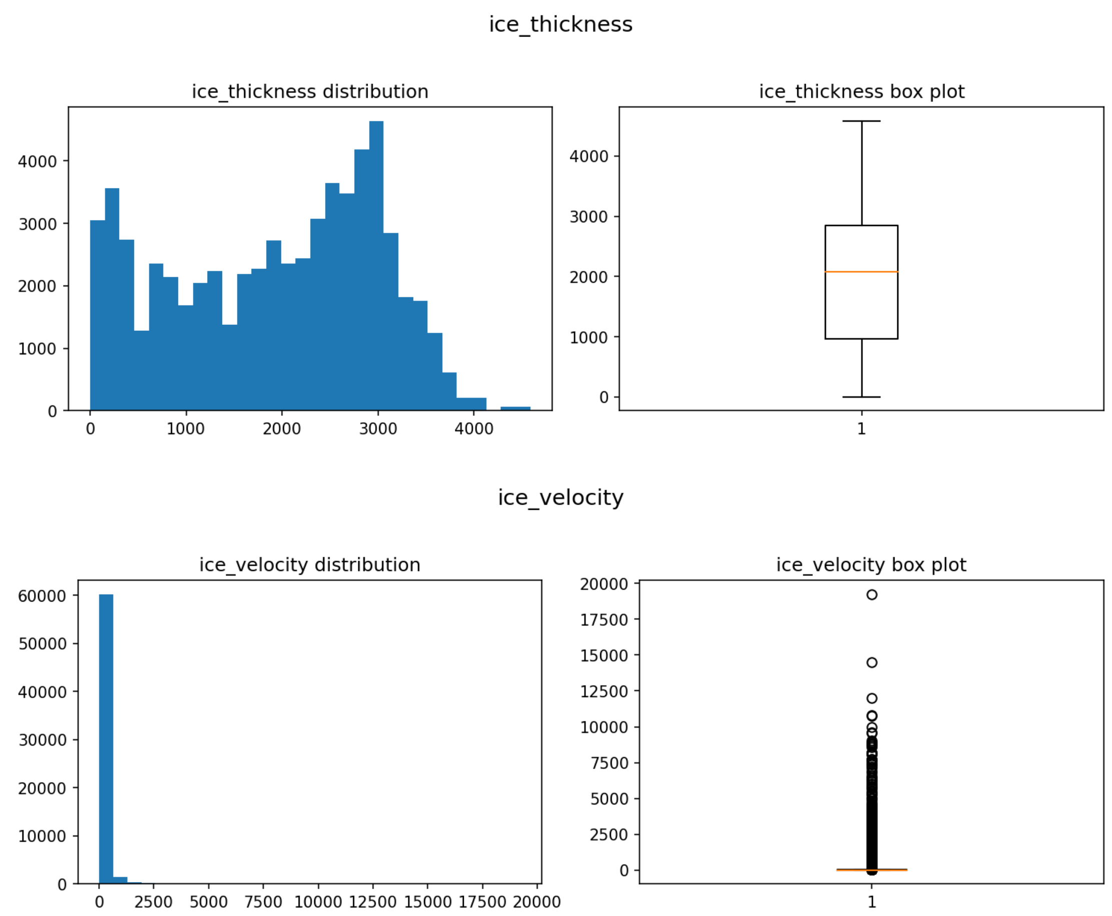


precipitation



ocean_temperature



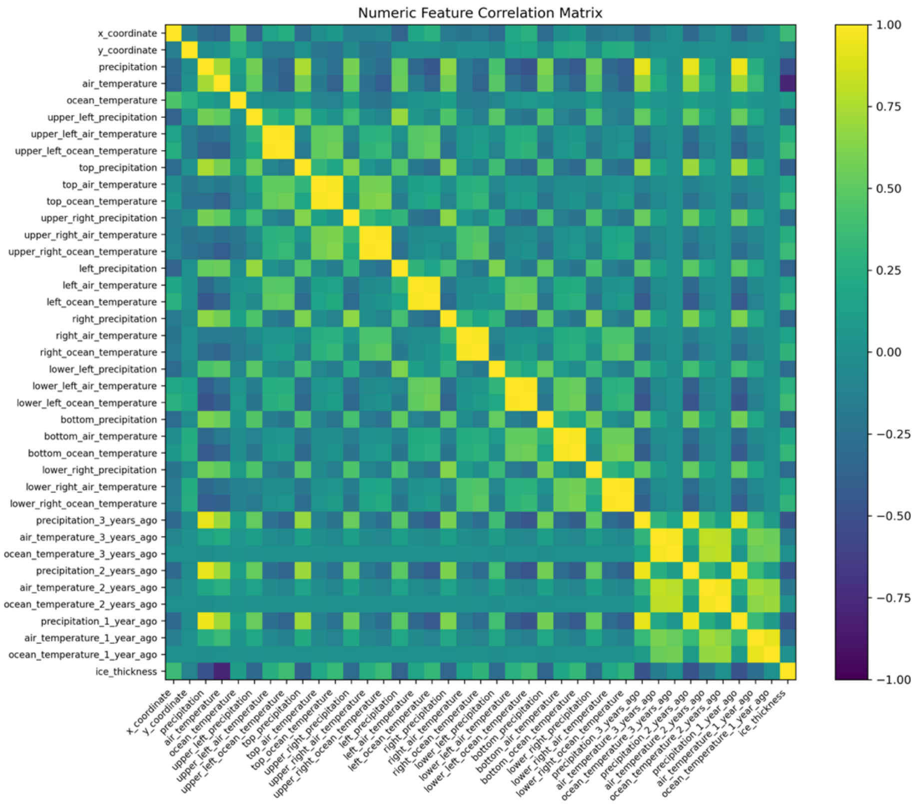


4.2.1 distribution plots from the Low-res-more-variables dataset

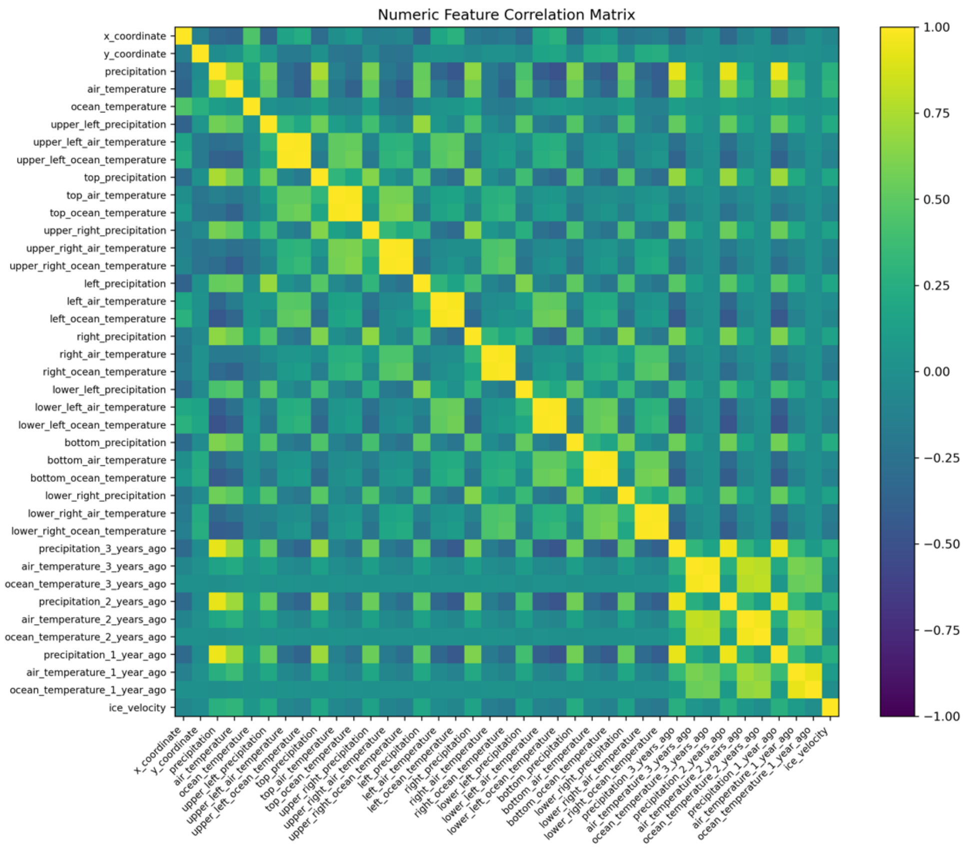
4.3. Correlation Heat Map

Compute the Pearson correlation coefficients between features and save the result as correlation_matrix.png in the corresponding directory.

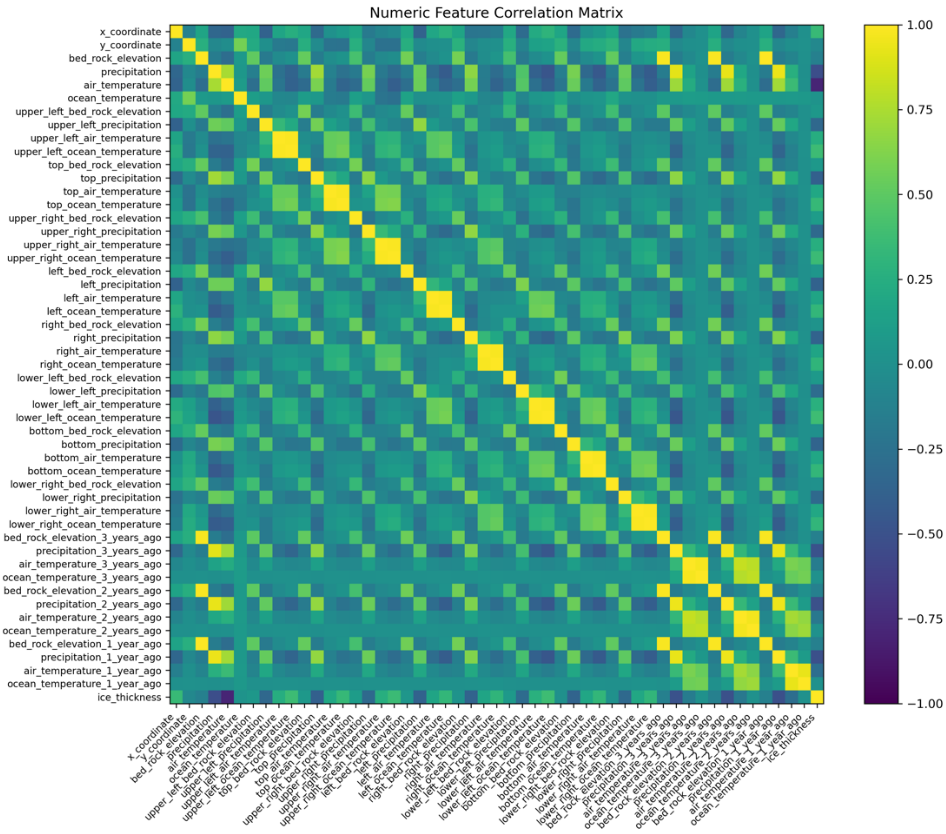
Below are the results for the Low-res-less-variables and Low-res-more-variables datasets using ice_thickness and ice_velocity as the labels, respectively:



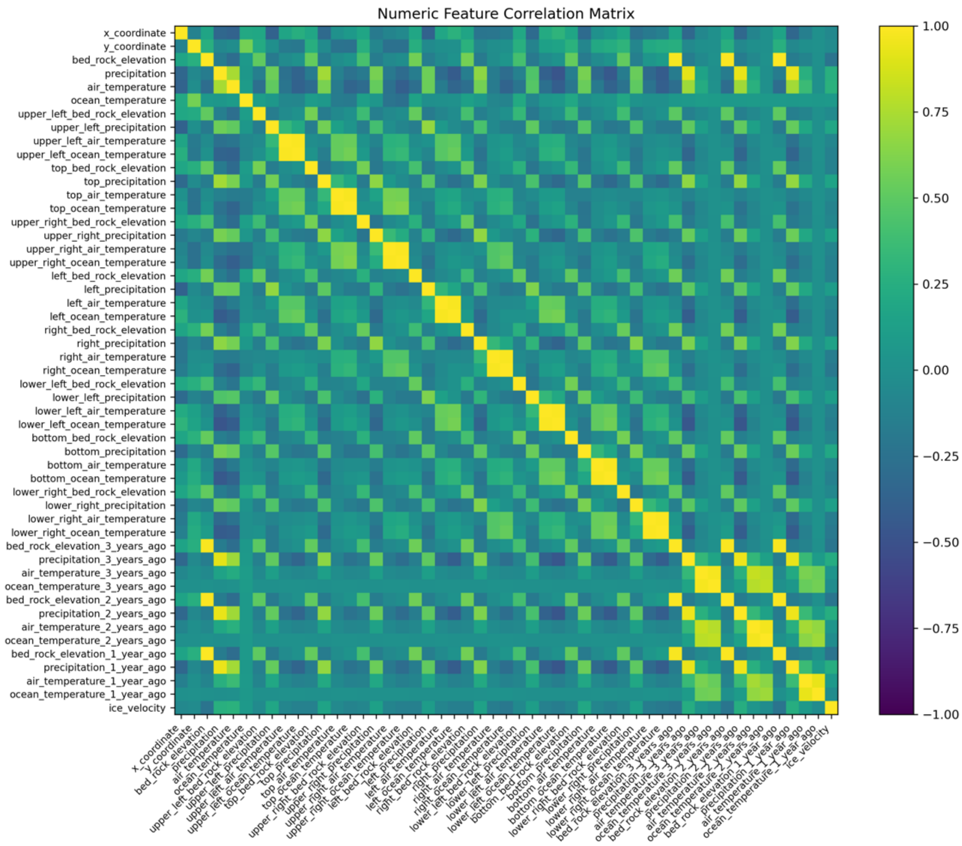
4.3.1 Low-res-less-variables dataset with ice_thickness as the label.



4.3.2 Low-res-less-variables dataset with ice_velocity as the label.



4.3.3 Low-res-more-variables dataset with ice_thickness as the label.



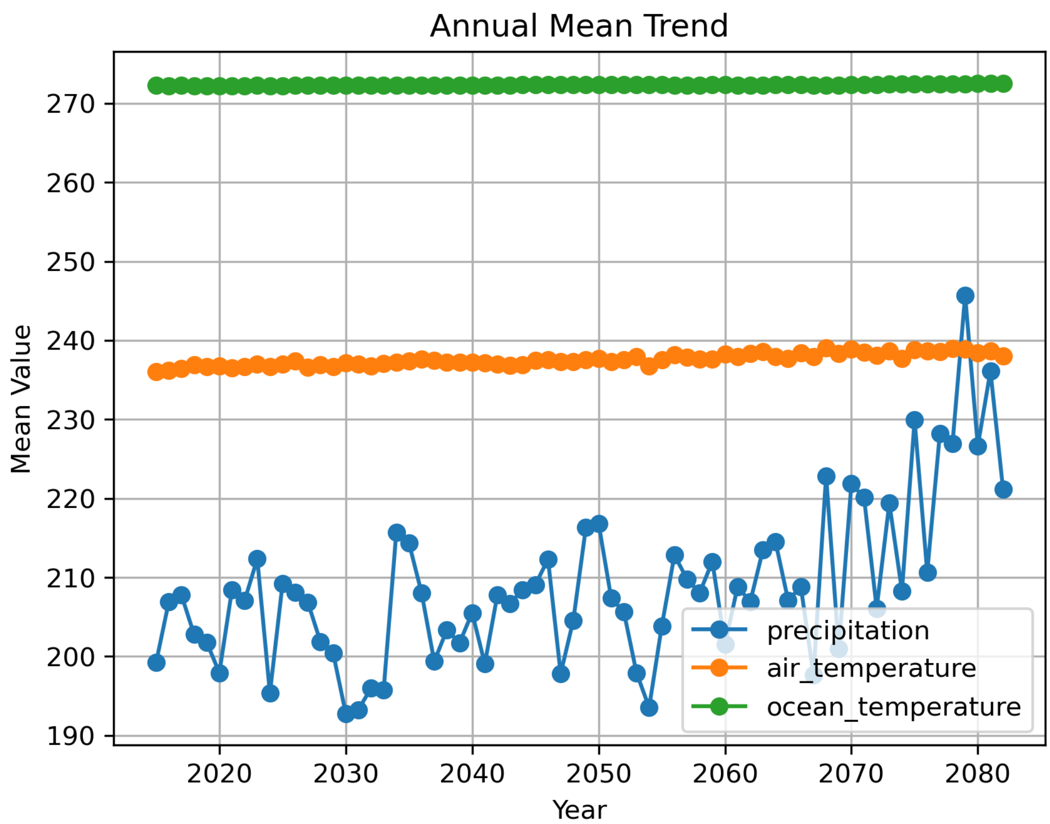
4.3.4 Low-res-more-variables dataset with ice_velocity as the label.

It can be observed that precipitation is highly positively correlated with precipitation from the preceding three years, and exhibits a strong negative correlation with ocean temperature at the eight surrounding locations. In future work, feature optimization or dimensionality reduction may be considered.

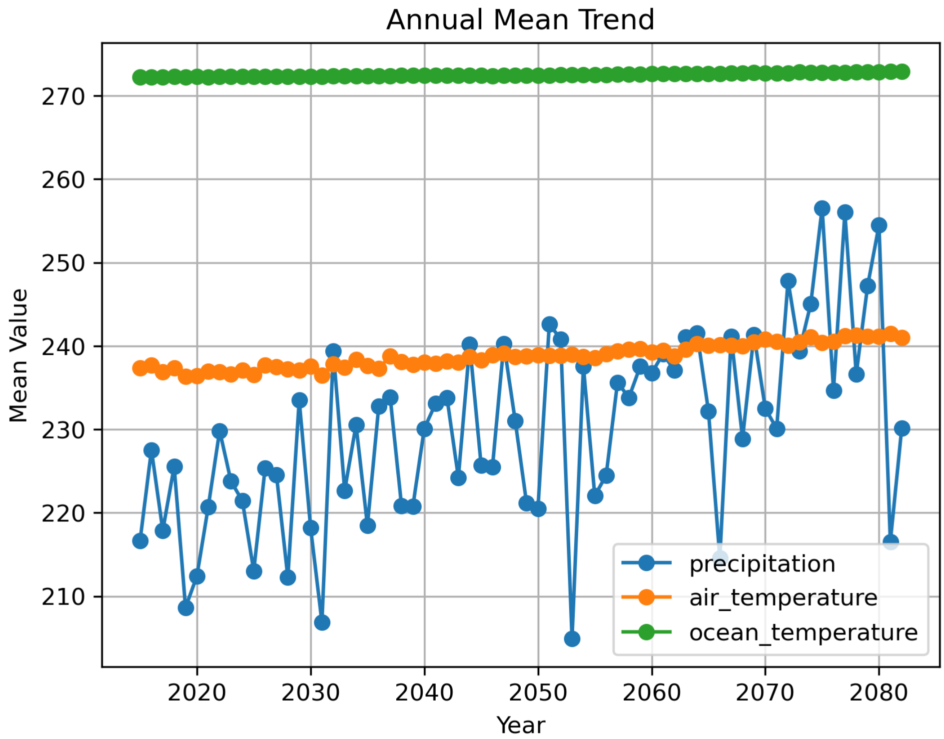
4.4. Time Trend

Plot the annual averages of “precipitation,” “air_temperature,” and “ocean_temperature,” and save the figure as “yearly_trend.png” in the corresponding dataset directory.

Below is an example from the Low-res-less-variables dataset:



Below is an example from the Low-res-more-variables dataset:

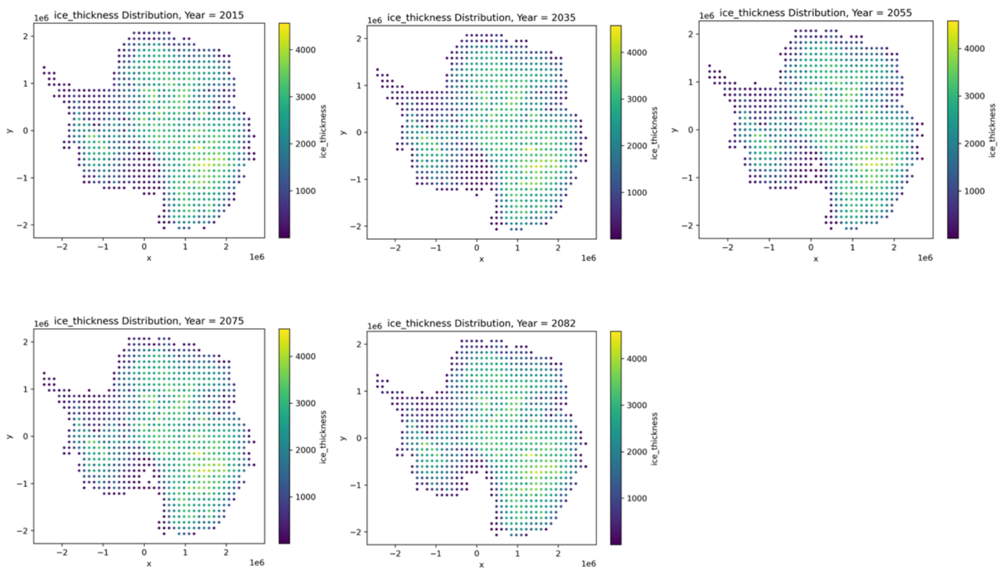


It can be observed that precipitation exhibits almost no change over successive years, whereas air temperature shows a slight upward trend. In contrast, ocean temperature increases with considerable fluctuations. Therefore, temporal information has a measurable impact on the predictions.

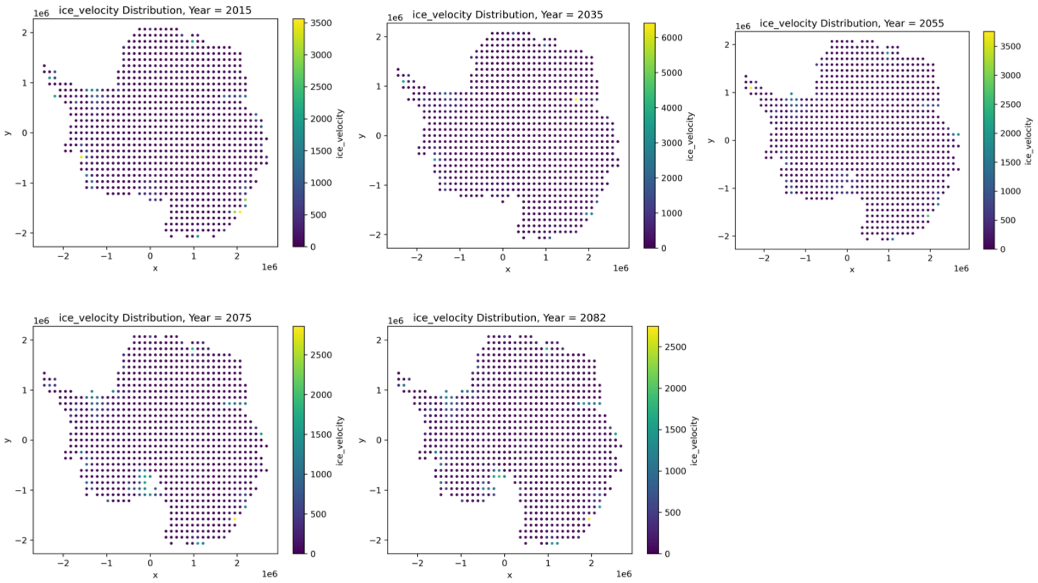
4.5. Geographic Distribution

Generate geographic distribution plots for each year and save them in the dataset's geographic_distribution directory.

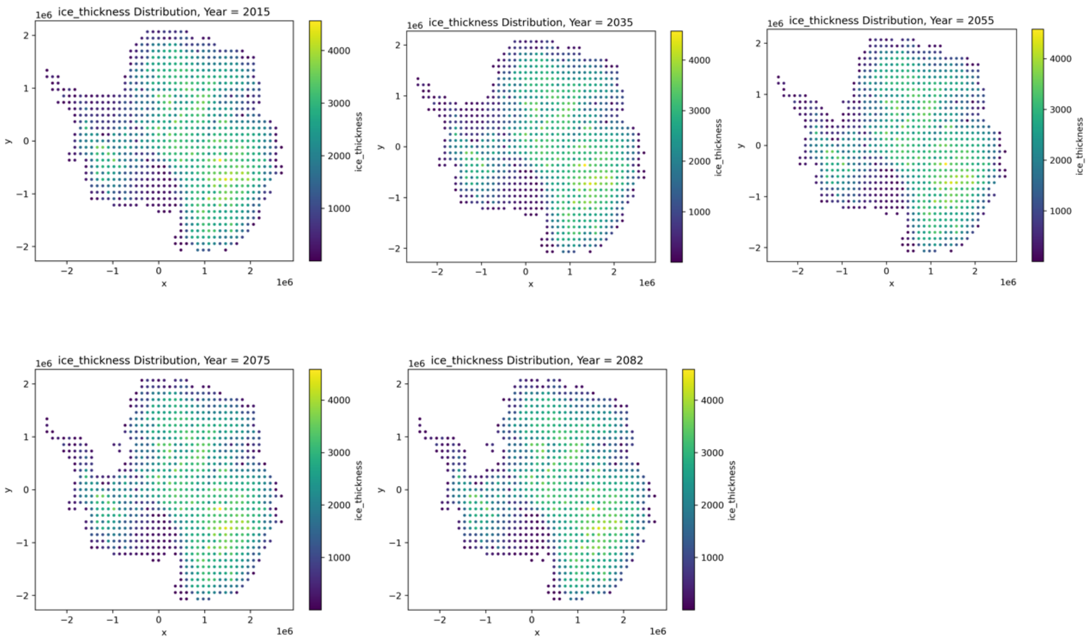
Below are the geographic distribution plots of several years from the Low-res-less-variables dataset, using ice_thickness as the label:



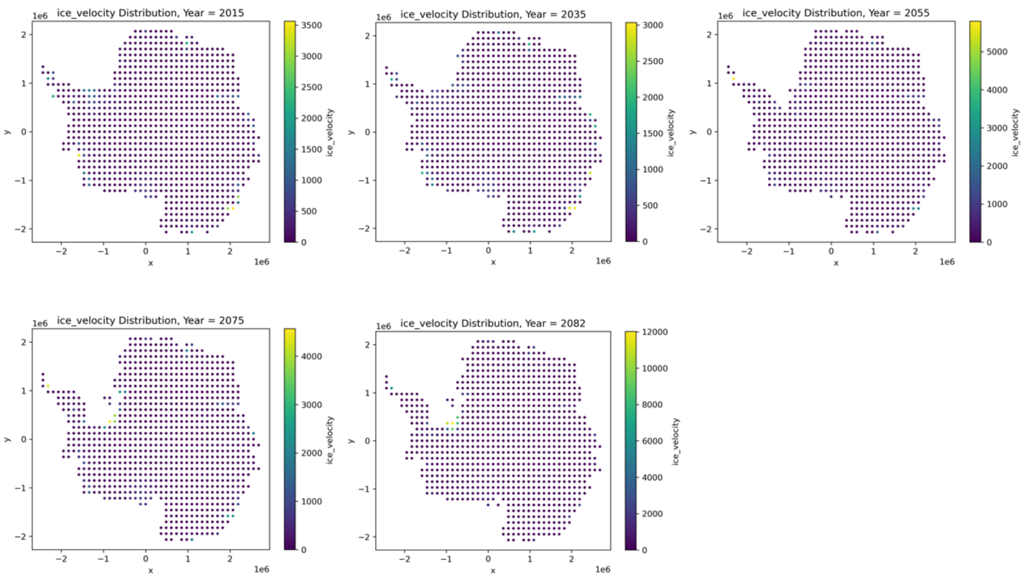
Below are the geographic distribution plots of several years from the Low-res-less-variables dataset, using ice_velocity as the label:



Below are the geographic distribution plots of several years from the Low-res-more-variables dataset, using ice_thickness as the label:



Below are the geographic distribution plots of several years from the Low-res-more-variables dataset, using ice_velocity as the label:



5. DATA PREPROCESSING

5.1. Outlier Handling

Outliers are clipped based on the interquartile range (IQR). For each numerical column feature, values are constrained to the interval [Q1 − 1.5·IQR, Q3 + 1.5·IQR], removing extreme noise and reducing the impact of outliers on model training.

5.2. Missing Value Imputation

Use scikit-learn's SimpleImputer to batch-impute missing values in the training and test sets, with the default strategy set to "median."

5.3. Standardization

Use scikit-learn's StandardScaler to perform z-score normalization on the numerical columns of the training and test sets.

5.4. Comparison Between Preprocessed Data and Raw Data

Below are the Preprocessed Basic Statistics results for the Low-res-less-variables and Low-res-more-variables datasets using ice_thickness and ice_velocity as the labels, respectively:

processed_data_describe_summary

	count	mean	std	min	25%	50%	75%	max
year	62605.0	-3.39943710909456E-15	1.0000079866782800	-1.7000453140003600	-0.8845947238644470	-0.018178471845038700	0.84823778017437	1.7146540321937800
x_coordinate	62605.0	-1.61164553115461E-17	1.0000079866781600	-2.4605222323302600	-0.8004836859857670	0.1332879963330090	0.8595548603587230	1.8970789518240300
y_coordinate	62605.0	-2.72390793997962E-18	1.0000079866780200	-2.1380212470264100	-0.8045801216021400	0.04397332184967100	0.7713048448083660	2.1047459702326400
precipitation	62605.0	-3.99506497863678E-17	1.0000079866781700	-1.6270917656937200	-0.7551256253550830	-0.4479385244964980	0.43155808644235200	2.2115836541385000
air_temperature	62605.0	-1.25753749895726E-15	1.000007986678220	-2.0306658323903800	-0.8685137051181530	0.003430816410878400	0.7373930966076210	3.089718282637520
ocean_temperature	62605.0	1.18181285822582E-14	1.0000079866782300	-2.151092403609360	-0.7513897596367710	-0.1564768177579160	0.5526072182755580	2.5086026851440000
upper_left_precipitation	62605.0	-5.99259746795516E-17	1.0000079866782000	-1.7904989529069100	-0.7405774782104880	-0.4242269276297300	0.44436216566604700	2.22177163148085
upper_left_air_temperature	62605.0	-1.16038478243132E-15	1.0000079866782600	-2.5037297238719000	-0.5600255563059790	0.05361517457467670	0.7357772220713020	2.6794813896372200
upper_left_ocean_temperature	62605.0	-6.12189229817286E-14	1.0000079866782000	-2.2137941440928900	-0.5432477169858480	-0.03478153005881020	0.570449901085542	2.2409963281926700
top_precipitation	62605.0	1.4164321287894E-16	1.0000079866782800	-1.7498771903374100	-0.7437234977379680	-0.4307557608434880	0.4467724717884680	2.232516426078120
top_air_temperature	62605.0	-1.48180591934891E-15	1.0000079866782400	-2.6400708584253800	-0.6098125333005980	0.04938069283182050	0.7436930167825910	2.289687474797000
top_ocean_temperature	62605.0	1.28437707185919E-13	1.0000079866782300	-2.2956187839589700	-0.5798365165828750	-0.06162522323387220	0.5640183283345210	2.279800595710620
upper_right_precipitation	62605.0	6.80976984994905E-17	1.000007986678310	-1.7901440933538700	-0.7403485613561450	-0.4238767760644310	0.44849350571570700	2.2317566063234900
upper_right_air_temperature	62605.0	1.63434476398777E-16	1.0000079866782100	-2.482212518761870	-0.5438733143704670	0.06272664223053450	0.7483528218904670	2.2594567965844700
upper_right_ocean_temperature	62605.0	7.89570114868759E-14	1.0000079866781800	-2.1917650761488600	-0.5337298077584140	-0.02555662766739580	0.571627037835219	2.2296623062256700
left_precipitation	62605.0	2.17912635198837E-17	1.0000079866782400	-1.7447934414382700	-0.7438891394308120	-0.4310593576051660	0.4340245462205510	2.2008950746976000
left_air_temperature	62605.0	4.5398465666327E-18	1.0000079866782100	-2.692006864581760	-0.6341782972326250	0.0481400580059810	0.7377074143334690	2.7955359816826100
left_ocean_temperature	62605.0	-1.78942592270394E-14	1.0000079866782200	-2.3446954302238700	-0.5961849493803500	-0.06199751883212520	0.5694887045153640	2.3179991853589800
right_precipitation	62605.0	-5.90180053662251E-17	1.0000079866781800	-1.7320104704275800	-0.7439547769703530	-0.4330050067715730	0.4371930484005200	2.208914786456830
right_air_temperature	62605.0	-2.65671821079346E-15	1.0000079866782400	-2.6981157450885600	-0.6346663323582740	0.04552126661835040	0.7409666094619180	2.397191258703540
right_ocean_temperature	62605.0	-3.35294908025225E-14	1.0000079866782000	-2.342506591242710	-0.5968182056768750	-0.06443980473088650	0.5669740513670440	2.312662436932970
lower_left_precipitation	62605.0	7.80853609460824E-17	1.0000079866781400	-1.7973274980723800	-0.7364483976791000	-0.4225878780430060	0.4278334315276810	2.174256175337850
lower_left_air_temperature	62605.0	-1.76146046785349E-16	1.000007986678240	-2.463911524577270	-0.5443193462422020	0.05795674420999630	0.735408772647844	2.6550009509829100
lower_left_ocean_temperature	62605.0	1.55226433806305E-14	1.000007986678280	-2.20153297676526	-0.5315260794330390	-0.01736711401147660	0.5818118521217750	2.251818749454000
bottom_precipitation	62605.0	-1.03508501719226E-16	1.000007986678210	-1.4539603362217800	-0.7410309499607400	-0.4297663396909850	0.42079602290506200	2.163536482203760
bottom_air_temperature	62605.0	2.95816402281787E-15	1.0000079866782400	-2.6244356683783600	-0.6102517188971350	0.05044313690451720	0.7325375807570180	2.746721530238240
bottom_ocean_temperature	62605.0	5.81463548254316E-14	1.0000079866782000	-2.3089359510083500	-0.5797268637119600	-0.05230796651310870	0.5730791944856320	2.30228828178202
lower_right_precipitation	62605.0	-1.04416471032552E-16	1.0000079866781400	-1.770138929464360	-0.7369291768228380	-0.42757607165714900	0.42306545482487400	2.163057402296440
lower_right_air_temperature	62605.0	1.86133709231941E-15	1.0000079866781800	-2.506369301897690	-0.5601585675801620	0.05578655337374110	0.7373152552981920	2.3323305186573800
lower_right_ocean_temperature	62605.0	-1.03461287314933E-13	1.000007986678150	-2.2248178485285300	-0.5431394140338170	-0.0309751405249917	0.5779795422959900	2.2596579767907000
precipitation_3_years_ago	62605.0	-3.26868952797554E-17	1.0000079866782300	-1.619857068774220	-0.7455298953844710	-0.44735177770130100	0.43245068719122700	2.199421561054770
air_temperature_3_years_ago	62605.0	2.75296295800607E-15	1.0000079866782200	-2.7289872281297600	-0.6632655469575430	0.05499900737760680	0.713882240490598	2.674757092433860
ocean_temperature_3_years_ago	62605.0	1.3594116559125E-14	1.0000079866782100	-2.37184544775017	-0.6102967880569570	-0.07569477192501420	0.5640689850718510	2.325617644765060
precipitation_2_years_ago	62605.0	2.25176389704982E-16	1.0000079866781600	-1.6201542915777600	-0.7486113720954770	-0.44625047140868000	0.4333611548342100	2.20631994522874
air_temperature_2_years_ago	62605.0	1.47091028758899E-15	1.0000079866782100	-2.887026270885070	-0.7234471827925520	0.03695043553448410	0.7189388759357920	2.7898431678329900
ocean_temperature_2_years_ago	62605.0	-7.38288008052076E-14	1.0000079866782200	-2.4716907913747100	-0.6530447522641570	-0.100337012895219	0.5593859404762130	2.3780319795867700
precipitation_1_year_ago	62605.0	-8.89809927060009E-17	1.0000079866782700	-1.6236009824978400	-0.7514532761920010	-0.4469427687601360	0.4328934190558460	2.2094134619276200
air_temperature_1_year_ago	62605.0	-5.79284421902332E-16	1.0000079866782200	-3.0618857832971300	-0.7885447257389620	0.02110001151121440	0.7270159792998190	2.9260002588930700
ocean_temperature_1_year_ago	62605.0	2.88153141277311E-14	1.000007986678230	-2.583813912912610	-0.7003960720656960	-0.12573462873126200	0.5552158218322470	2.43863366267916
ice_thickness	62605.0	1918.8392251958700	1071.6295773098800	0.0100353881717	982.284790039	2073.45605469	2837.94873047	4586.17773438

5.4.1 Preprocessed Low-res-less-variables dataset with ice_thickness as the label.

processed_data_describe_summary

	count	mean	std	min	25%	50%	75%	max
year	62605.0	-3.39943710909456E-15	1.0000079866782800	-1.7000453140003600	-0.8845947238644470	-0.018178471845038700	0.84823778017437	1.7146540321937800
x_coordinate	62605.0	-1.61164553115461E-17	1.0000079866781600	-2.4605222323302600	-0.8004836859857670	0.1332879963330090	0.8595548603587230	1.8970789518240300
y_coordinate	62605.0	-2.72390793997962E-18	1.0000079866780200	-2.1380212470264100	-0.8045801216021400	0.043973321849671100	0.7713048448083660	2.1047459702326400
precipitation	62605.0	-3.99506497863678E-17	1.0000079866781700	-1.6270917656937200	-0.7551256253550830	-0.4479385244964980	0.43155808644235200	2.2115836541385000
air_temperature	62605.0	-1.25753749895726E-15	1.000007986678220	-2.0306658323903800	-0.8685137051181530	0.003430816410878400	0.7373930966076210	3.089718282637520
ocean_temperature	62605.0	1.18181285822582E-14	1.0000079866782300	-2.151092403609360	-0.7513897596367710	-0.15647681775791600	0.5526072182755580	2.5086026851440000
upper_left_precipitation	62605.0	-5.99259746795516E-17	1.0000079866782000	-1.7904989529069100	-0.7405774782104880	-0.4242269276297300	0.44436216566604700	2.221771631480850
upper_left_air_temperature	62605.0	-1.16038478243132E-15	1.0000079866782600	-2.5037297238719000	-0.5600255563059790	0.05361517457467670	0.7357772220713020	2.6794813896372200
upper_left_ocean_temperature	62605.0	-6.12189229817286E-14	1.0000079866782000	-2.2137941440928900	-0.5432477169858480	-0.03478153005881020	0.570449901085542	2.2409963281926700
top_precipitation	62605.0	1.4164321287894E-16	1.0000079866782800	-1.7498771903374100	-0.7437234977379680	-0.4307557608434880	0.4467724717884680	2.232516426078120
top_air_temperature	62605.0	-1.48180591934891E-15	1.0000079866782400	-2.6400708584253800	-0.6098125333005980	0.04938069283182050	0.7436930167825910	2.289687474797000
top_ocean_temperature	62605.0	1.28437707185919E-13	1.0000079866782300	-2.2956187839589700	-0.5798365165828750	-0.06162522323387220	0.5640183283345210	2.279800595710620
upper_right_precipitation	62605.0	6.80976984994905E-17	1.000007986678310	-1.7901440933538700	-0.7403485613561450	-0.4238767760644310	0.44849350571570700	2.2317566063234900
upper_right_air_temperature	62605.0	1.63434476398777E-16	1.0000079866782100	-2.482212518761870	-0.5438733143704670	0.06272664223053450	0.7483528218904670	2.2594567965844700
upper_right_ocean_temperature	62605.0	7.89570114868759E-14	1.0000079866781800	-2.1917650761488600	-0.5337298077584140	-0.02555662766739580	0.571627037835219	2.2296623062256700
left_precipitation	62605.0	2.1791263519837E-17	1.0000079866782400	-1.7447934414382700	-0.7438891394308120	-0.4310593576051660	0.4340245462205510	2.2008950746976000
left_air_temperature	62605.0	4.5398465666327E-18	1.0000079866782100	-2.692006864581760	-0.6341782972326250	0.04814005800598100	0.7377074143334690	2.7955359816826100
left_ocean_temperature	62605.0	-1.78942592270394E-14	1.0000079866782200	-2.3446954302238700	-0.5961849493803500	-0.06199751883212520	0.5694887045153640	2.3179991853589800
right_precipitation	62605.0	-5.90180053662251E-17	1.0000079866781800	-1.7320104704275800	-0.7439547769703530	-0.4330050067715730	0.4371930484005200	2.208914786456830
right_air_temperature	62605.0	-2.65671821079346E-15	1.0000079866782400	-2.6981157450885600	-0.6346663323582740	0.045521266618350400	0.7409666094619180	2.397191258703540
right_ocean_temperature	62605.0	-3.35294908025225E-14	1.0000079866782000	-2.342506591242710	-0.5968182056768750	-0.06443980473088650	0.5669740513670440	2.312662436932970
lower_left_precipitation	62605.0	7.80853609460824E-17	1.0000079866781400	-1.7973274980723800	-0.7364483976791000	-0.42258787804300600	0.4278334315276810	2.174256175337850
lower_left_air_temperature	62605.0	-1.76146046785349E-16	1.000007986678240	-2.463911524577270	-0.5443193462422020	0.057956744209996300	0.735408772647844	2.6550009509829100
lower_left_ocean_temperature	62605.0	1.55226433806305E-14	1.000007986678280	-2.20153297676526	-0.5315260794330390	-0.01736711401147660	0.5818118521217750	2.251818749454000
bottom_precipitation	62605.0	-1.03508501719226E-16	1.000007986678210	-1.4539603362217800	-0.7410309499607400	-0.4297663396909850	0.42079602290506200	2.163536482203760
bottom_air_temperature	62605.0	2.95816402281787E-15	1.0000079866782400	-2.6244356683783600	-0.6102517188971350	0.05044313690451720	0.7325375807570180	2.746721530238240
bottom_ocean_temperature	62605.0	5.81463548254316E-14	1.0000079866782000	-2.3089359510083500	-0.5797268637119600	-0.05230796651310870	0.5730791944856320	2.30228828178202
lower_right_precipitation	62605.0	-1.04416471032552E-16	1.0000079866781400	-1.770138929464360	-0.7369291768228380	-0.42757607165714900	0.42306545482487400	2.163057402296440
lower_right_air_temperature	62605.0	1.86133709231941E-15	1.0000079866781800	-2.506369301897690	-0.5601585675801620	0.05578655337374110	0.7373152552981920	2.3323305186573800
lower_right_ocean_temperature	62605.0	-1.03461287314933E-13	1.000007986678150	-2.2248178485285300	-0.5431394140338170	-0.0309751405249917	0.5779795422959900	2.2596579767907000
precipitation_3_years_ago	62605.0	-3.26868952797554E-17	1.0000079866782300	-1.619857068774220	-0.7455298953844710	-0.44735177770130100	0.43245068719122700	2.199421561054770
air_temperature_3_years_ago	62605.0	2.75296295800607E-15	1.0000079866782200	-2.7289872281297600	-0.6632655469575430	0.05499900737760680	0.713882240490598	2.674757092433860
ocean_temperature_3_years_ago	62605.0	1.3594116559125E-14	1.0000079866782100	-2.37184544775017	-0.6102967880569570	-0.07569477192501420	0.5640689850718510	2.325617644765060
precipitation_2_years_ago	62605.0	2.25176389704982E-16	1.0000079866781600	-1.6201542915777600	-0.7486113720954770	-0.44625047140868000	0.4333611548342100	2.20631994522874
air_temperature_2_years_ago	62605.0	1.47091028758899E-15	1.0000079866782100	-2.887026270885070	-0.7234471827925520	0.036950435534484100	0.7189388759357920	2.7898431678329900
ocean_temperature_2_years_ago	62605.0	-7.38288008052076E-14	1.0000079866782200	-2.4716907913747100	-0.6530447522641570	-0.100337012895219	0.5593859404762130	2.3780319795867700
precipitation_1_year_ago	62605.0	-8.89809927060009E-17	1.0000079866782700	-1.6236009824978400	-0.7514532761920010	-0.4469427687601360	0.4328934190558460	2.2094134619276200
air_temperature_1_year_ago	62605.0	-5.79284421902332E-16	1.0000079866782200	-3.0618857832971300	-0.7885447257389620	0.02110001151121440	0.7270159792998190	2.9260002588930700
ocean_temperature_1_year_ago	62605.0	2.88153141277311E-14	1.000007986678230	-2.583813912912610	-0.7003960720656960	-0.12573462873126200	0.5552158218322470	2.43863366267916
ice_velocity	62605.0	86.09490362408030	308.1544644910150	0.0	2.61102294922	8.17558765411	27.8806819916	13779.2851562

5.4.2 Preprocessed Low-res-less-variables dataset with ice_velocity as the label.

processed_data_describe_summary

	count	mean	std	min	25%	50%	75%	max
year	62331.0	6.8579440380112E-16	1.0000080217872100	-1.6945511303903200	-0.8776259364232310	-0.00964291783320094	0.8583401007568290	1.7263231193468600
x_coordinate	62331.0	3.00947012306343E-17	1.0000080217871800	-2.515992125634160	-0.7289080090135230	0.11207275174912900	0.8479309174164490	1.8991568683697600
y_coordinate	62331.0	4.24061699158937E-17	1.000008021787250	-2.0979794434773900	-0.7735531483121330	-0.051138805494719600	0.7916779277922630	1.9957018324879500
bed_rock_elevation	62331.0	1.55033309369934E-17	1.0000080217871800	-2.596433555004980	-0.6623681531535510	-0.027372170689173100	0.6270087814140650	2.561074183265490
precipitation	62331.0	1.91511735104036E-16	1.0000080217872300	-2.5553252293125700	-0.7611945473856690	-0.4327687732228690	0.43489257389893100	2.229023255825830
air_temperature	62331.0	1.18190099378491E-15	1.0000080217871700	-2.0498739245774500	-0.8549824213897040	0.002359894399399570	0.7585517532798680	3.1788530152842200
ocean_temperature	62331.0	1.08282558949109E-13	1.0000080217871700	-2.2765883803752900	-0.7687497727274230	-0.20371751332752100	0.5863478409278870	2.618994261410850
upper_left_bed_rock_elevation	62331.0	2.18870554404613E-17	1.0000080217872100	-2.5569918388350800	-0.6557880108525330	-0.03073839727071490	0.6116812078025010	2.512885035785050
upper_left_precipitation	62331.0	7.02209695381466E-17	1.0000080217871900	-2.5262316949645100	-0.7389044403228160	-0.4126822488431910	0.4526470627716460	2.2399743174133400
upper_left_air_temperature	62331.0	3.40890888485185E-15	1.0000080217872100	-2.5030356152323500	-0.5563842920905540	0.05772748275175920	0.74138325667048	2.688034579812450
upper_left_ocean_temperature	62331.0	2.14639057019457E-14	1.000008021787130	-2.2347205068414000	-0.5398554313999000	-0.07358166529534820	0.5900546188944700	2.28491969433593
top_bed_rock_elevation	62331.0	-1.6415291580346E-17	1.0000080217871800	-2.584307590666600	-0.6584519939918840	-0.010372858829363500	0.6254517371245950	2.5513073337993100
top_precipitation	62331.0	-6.11013631046211E-17	1.000008021787180	-2.53312177433808	-0.7443020774140030	-0.4155493600440580	0.44824438720204900	2.2370640841261300
top_air_temperature	62331.0	-7.24096750821928E-16	1.0000080217871900	-2.63878747147929	-0.6052096401703920	0.0449591763434636	0.7505089140355390	2.355717469734110
top_ocean_temperature	62331.0	7.8209744773915E-15	1.000008021787180	-2.3280255970766900	-0.5787007674502930	-0.1000928202322780	0.5875157856339480	2.3368406152602700
upper_right_bed_rock_elevation	62331.0	-1.09435277202306E-17	1.000008021787160	-2.5866233769948600	-0.659549079001352	-0.008931605255562440	0.6251671196609840	2.552241417654490
upper_right_precipitation	62331.0	4.01262683075124E-17	1.0000080217871400	-2.526280571898760	-0.7384748215991540	-0.4150640041143850	0.4533956786005830	2.24120142890019
upper_right_air_temperature	62331.0	1.18554883635832E-17	1.0000080217871100	-2.4731992769638100	-0.5363748502547020	0.06710194225221280	0.7548414342180380	2.255127004438410
upper_right_ocean_temperature	62331.0	-3.77296357367818E-14	1.0000080217871600	-2.2097362500943900	-0.528194290422864	-0.06332031289429010	0.59283368269155	2.274375642363120
left_bed_rock_elevation	62331.0	3.51104847690733E-17	1.0000080217871600	-2.5890144667242200	-0.6629149267380720	-0.012320312355174000	0.6211514332526920	2.547250973238840
left_precipitation	62331.0	1.39529978432941E-16	1.0000080217871400	-2.5222771482424000	-0.7453214825894270	-0.4191143677193120	0.4393156278458900	2.216271293498870
left_air_temperature	62331.0	-3.97614840501713E-16	1.0000080217871900	-2.6705545891985600	-0.6222987219445590	0.04079969789957100	0.7432051895581080	2.791461056812110
left_ocean_temperature	62331.0	-4.29205157187446E-14	1.0000080217871700	-2.357663217750130	-0.591802921025236	-0.10352754852176200	0.5854372767914240	2.351297573516350
right_bed_rock_elevation	62331.0	-3.64784257341021E-17	1.0000080217871800	-2.5894225790359500	-0.6574371091995930	-0.007894836877349890	0.6305532040246440	2.562538673861000
right_precipitation	62331.0	8.20764579017298E-17	1.0000080217871200	-2.526945690009100	-0.7465247386781280	-0.4199787887312230	0.4404225622091860	2.2208435135401600
right_air_temperature	62331.0	1.58681151943344E-16	1.000008021787190	-2.6805068880300900	-0.6224625658814940	0.03965526989490550	0.7495669822175730	2.329697202531770
right_ocean_temperature	62331.0	-5.00994699032159E-14	1.000008021787170	-2.371711747382110	-0.594174549769012	-0.10299234762797300	0.5908502486396950	2.3683874462527200
lower_left_bed_rock_elevation	62331.0	-3.28305831606919E-17	1.0000080217872500	-2.5679435577541400	-0.6603320207315990	0.004238895586182930	0.6114090039500980	2.519020540972640
lower_left_precipitation	62331.0	1.62328994516755E-16	1.000008021787190	-2.4948377131793900	-0.7348455888420200	-0.4164012269146990	0.4384824940495610	2.1984746183869300
lower_left_air_temperature	62331.0	4.90634826123674E-16	1.0000080217871200	-2.4515045976083100	-0.5378354606587170	0.06103738641198180	0.737943963974345	2.651613100923940
lower_left_ocean_temperature	62331.0	-2.61331441959108E-14	1.0000080217871700	-2.2060835516189500	-0.5254134537159790	-0.05906029724302240	0.5950332782193100	2.2757033761222000
bottom_bed_rock_elevation	62331.0	3.64784257341021E-18	1.0000080217871700	-2.6021676690551300	-0.6660408087618770	-0.009962264691206770	0.6247104314336230	2.560837291726870
bottom_precipitation	62331.0	-7.29568514682043E-18	1.0000080217870700	-2.4987477314825500	-0.7416578651815840	-0.4198599664106990	0.4297353790190580	2.18682524532002
bottom_air_temperature	62331.0	-1.51020682539183E-15	1.0000080217872000	-2.6165763817393700	-0.6057639359408080	0.040473061051873700	0.7347776945915630	2.7455901403901200
bottom_ocean_temperature	62331.0	9.55005185718794E-15	1.0000080217872200	-2.3320956334732000	-0.5781125255816680	-0.09422263753145630	0.5912095463461130	2.345192654237770
lower_right_bed_rock_elevation	62331.0	-1.55033309369934E-17	1.000008021787140	-2.602127672623460	-0.6626741297007140	-0.006833916642421470	0.6302948989144490	2.569748441837190
lower_right_precipitation	62331.0	-5.65415598878583E-17	1.0000080217872500	-2.4944369819078600	-0.7361598346623910	-0.4201849998285460	0.4360249301679190	2.1943020774133800
lower_right_air_temperature	62331.0	-4.88810904836969E-16	1.0000080217872100	-2.502341428591040	-0.5561356297743020	0.05500843837134230	0.741334902770191	2.3722822622593700
lower_right_ocean_temperature	62331.0	6.56465749510902E-14	1.0000080217871800	-2.2449609704576200	-0.5408241743883990	-0.07260436728837180	0.5952670229910180	2.2994038190602000
bed_rock_elevation_3_years_ago	62331.0	1.3223429328612E-17	1.0000080217871700	-2.5823734119320700	-0.6558329831619930	-0.004745535166188570	0.62852730268472 3	2.5550677314548000
precipitation_3_years_ago	62331.0	-3.64784257341021E-18	1.0000080217872300	-2.522424548138100	-0.7488994919322910	-0.4291017335558970	0.433450545538249	2.2069756017440600
air_temperature_3_years_ago	62331.0	-2.26531023808774E-15	1.0000080217871800	-2.724119610699840	-0.65280622222167	0.04986512430365110	0.7280693700971070	2.7869360102044900
ocean_temperature_3_years_ago	62331.0	3.03281631553325E-14	1.0000080217871900	-2.405951951336530	-0.6107811037733710	-0.11315851290870600	0.58599946126871	2.3811703088317900
bed_rock_elevation_2_years_ago	62331.0	1.82392128670511E-18	1.0000080217871900	-2.5950375800454400	-0.6605617566565560	-0.004596514653163650	0.6290887922693660	2.5635646156582500
precipitation_2_years_ago	62331.0	-1.51385466796524E-16	1.0000080217872300	-2.5328917757462400	-0.7523007231742730	-0.43119875842186500	0.43475997854037000	2.2153510311123300
air_temperature_2_years_ago	62331.0	1.05057866114214E-15	1.0000080217871700	-2.8839874039041800	-0.7125507228138930	0.034745209842898700	0.7350737312463010	2.905868285217880
ocean_temperature_2_years_ago	62331.0	-5.26675510748967E-14	1.000008021787210	-2.5214430749738900	-0.656820506889562	-0.14072932443144500	0.5862612051666270	2.4508837732508700
bed_rock_elevation_1_year_ago	62331.0	1.27674490069358E-17	1.0000080217871700	-2.592970547093830	-0.6598429101527830	-0.006439273363259820	0.628908847807915	2.5620364847489600
precipitation_1_year_ago	62331.0	-2.37109767271664E-17	1.000008021787120	-2.5433372351308100	-0.7567283935567540	-0.43174123765272600	0.4343441674926200	2.2209530090666800
air_temperature_1_year_ago	62331.0	1.31322332642768E-16	1.000008021787180	-3.065001645315100	-0.7787040319839740	0.019247880847550300	0.7454943769034390	3.031791990234560
ocean_temperature_1_year_ago	62331.0	-2.61185528256171E-14	1.000008021787180	-2.6493371272234200	-0.7083617823230990	-0.17093850307052000	0.5856217809438140	2.5265971258442300
ice_thickness	62331.0	1917.8634570688900	1077.8319296910100	0.0104743046686	967.819152832	2080.35620117	2843.658569335	4586.45166016

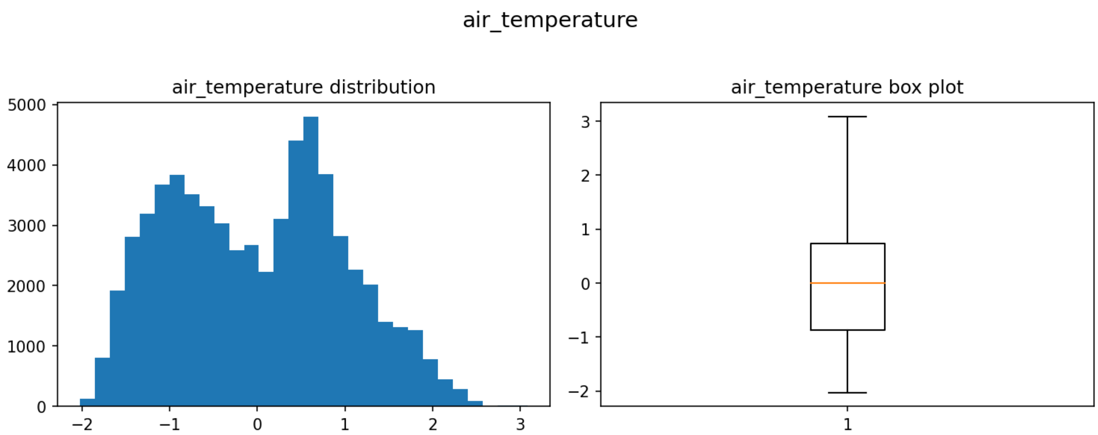
5.4.3 Preprocessed Low-res-more-variables dataset with ice_thickness as the label.

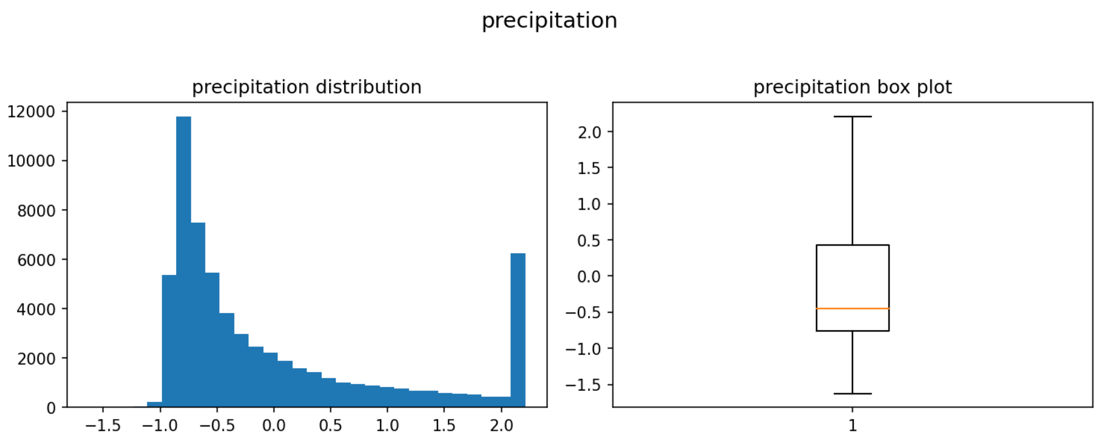
processed_data_describe_summary

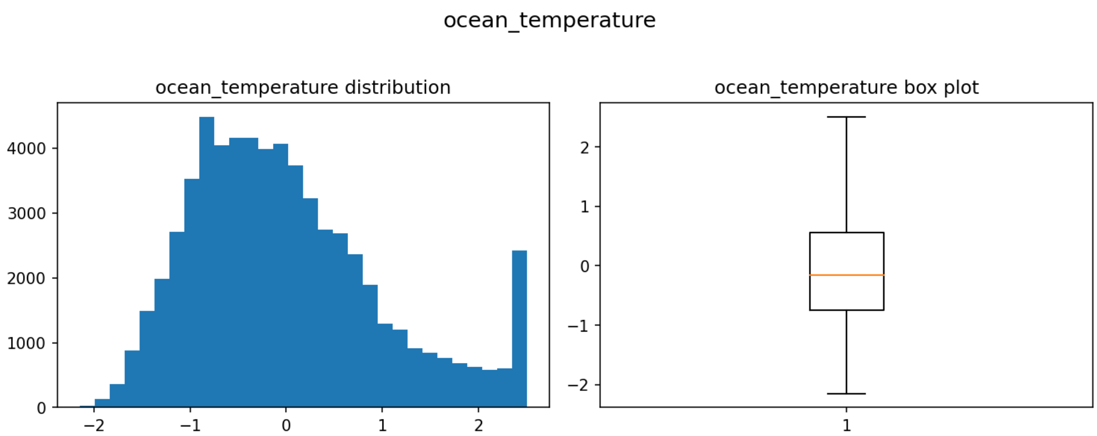
	count	mean	std	min	25%	50%	75%	max
year	62331.0	6.8579440380112E-16	1.0000080217872100	-1.6945511303903200	-0.8776259364232310	-0.00964291783320094	0.8583401007568290	1.7263231193468600
x_coordinate	62331.0	3.00947012306343E-17	1.0000080217871800	-2.515992125634160	-0.7289080090135230	0.11207275174912900	0.8479309174164490	1.8991568683697600
y_coordinate	62331.0	4.24061699158937E-17	1.000008021787250	-2.0979794434773900	-0.7735531483121330	-0.05113880549471960	0.7916779277922630	1.9957018324879500
bed_rock_elevation	62331.0	1.55033309369934E-17	1.0000080217871800	-2.596433555004980	-0.6623681531535510	-0.027372170689173100	0.6270087814140650	2.561074183265490
precipitation	62331.0	1.91511735104036E-16	1.0000080217872300	-2.5553252293125700	-0.7611945473856690	-0.4327687732228690	0.43489257389893100	2.229023255825830
air_temperature	62331.0	1.18190099378491E-15	1.0000080217871700	-2.0498739245774500	-0.8549824213897040	0.00235989439939570	0.7585517532798680	3.1788530152842200
ocean_temperature	62331.0	1.08282558949109E-13	1.0000080217871700	-2.2765883803752900	-0.7687497727274230	-0.20371751332752100	0.5863478409278870	2.618994261410850
upper_left_bed_rock_elevation	62331.0	2.18870554404613E-17	1.0000080217872100	-2.5569918388350800	-0.6557880108525330	-0.03073839727071490	0.6116812078025010	2.512885035785050
upper_left_precipitation	62331.0	7.02209695381466E-17	1.0000080217871900	-2.5262316949645100	-0.7389044403228160	-0.4126822488431910	0.4526470627716460	2.2399743174133400
upper_left_air_temperature	62331.0	3.40890888485185E-15	1.0000080217872100	-2.5030356152323500	-0.5563842920905540	0.05772748275175920	0.741383256670648	2.688034579812450
upper_left_ocean_temperature	62331.0	2.14639057019457E-14	1.000008021787130	-2.2347205068414000	-0.5398554313999000	-0.07358166529534820	0.5900546188944700	2.28491969433593
top_bed_rock_elevation	62331.0	-1.6415291580346E-17	1.0000080217871800	-2.584307590666600	-0.6584519939918840	-0.010372858829363500	0.6254517371245950	2.5513073337993100
top_precipitation	62331.0	-6.11013631046211E-17	1.000008021787180	-2.53312177433808	-0.7443020774140030	-0.4155493600440580	0.44824438720204900	2.2370640841261300
top_air_temperature	62331.0	-7.24096750821928E-16	1.0000080217871900	-2.63878747147929	-0.6052096401703920	0.0449591763434636	0.7505089140355390	2.355717469734110
top_ocean_temperature	62331.0	7.8209744773915E-15	1.000008021787180	-2.3280255970766900	-0.5787007674502930	-0.1000928202322780	0.5875157856339480	2.3368406152602700
upper_right_bed_rock_elevation	62331.0	-1.09435277202306E-17	1.000008021787160	-2.5866233769948600	-0.659549079001352	-0.008931605255562440	0.6251671196609840	2.552241417654490
upper_right_precipitation	62331.0	4.01262683075124E-17	1.0000080217871400	-2.526280571898760	-0.7384748215991540	-0.4150640041143850	0.4533956786005830	2.24120142890019
upper_right_air_temperature	62331.0	1.18554883635832E-17	1.0000080217871100	-2.4731992769638100	-0.5363748502547020	0.06710194225221280	0.7548414342180380	2.255127004438410
upper_right_ocean_temperature	62331.0	-3.77296357367818E-14	1.0000080217871600	-2.2097362500943900	-0.528194290422864	-0.06332031289429010	0.59283368269155	2.274375642363120
left_bed_rock_elevation	62331.0	3.51104847690733E-17	1.0000080217871600	-2.5890144667242200	-0.6629149267380720	-0.012320312355174000	0.6211514332526920	2.547250973238840
left_precipitation	62331.0	1.39529978432941E-16	1.0000080217871400	-2.5222771482424000	-0.7453214825894270	-0.4191143677193120	0.4393156278458900	2.216271293498870
left_air_temperature	62331.0	-3.97614840501713E-16	1.0000080217871900	-2.6705545891985600	-0.6222987219445590	0.04079969789957100	0.7432051895581080	2.791461056812110
left_ocean_temperature	62331.0	-4.29205157187446E-14	1.0000080217871700	-2.357663217750130	-0.591802921025236	-0.10352754852176200	0.5854372767914240	2.351297573516350
right_bed_rock_elevation	62331.0	-3.64784257341021E-17	1.0000080217871800	-2.5894225790359500	-0.6574371091995930	-0.007894836877349890	0.6305532040246440	2.562538673861000
right_precipitation	62331.0	8.20764579017298E-17	1.0000080217871200	-2.526945690009100	-0.7465247386781280	-0.4199787887312230	0.4404225622091860	2.2208435135401600
right_air_temperature	62331.0	1.58681151943344E-16	1.000008021787190	-2.6805068880300900	-0.6224625658814940	0.03965526989490550	0.7495669822175730	2.329697202531770
right_ocean_temperature	62331.0	-5.00994699032159E-14	1.000008021787170	-2.371711747382110	-0.594174549769012	-0.10299234762797300	0.5908502486396950	2.3683874462527200
lower_left_bed_rock_elevation	62331.0	-3.28305831606919E-17	1.0000080217872500	-2.5679435577541400	-0.6603320207315990	0.004238895586182930	0.6114090039500980	2.519020540972640
lower_left_precipitation	62331.0	1.62328994516755E-16	1.000008021787190	-2.4948377131793900	-0.7348455888420200	-0.41640122691469900	0.4384824940495610	2.1984746183869300
lower_left_air_temperature	62331.0	4.90634826123674E-16	1.0000080217871200	-2.4515045976083100	-0.5378354606587170	0.06103738641198180	0.737943963974345	2.651613100923940
lower_left_ocean_temperature	62331.0	-2.61331441959108E-14	1.0000080217871700	-2.2060835516189500	-0.5254134537159790	-0.05906029724302240	0.5950332782193100	2.2757033761222000
bottom_bed_rock_elevation	62331.0	3.64784257341021E-18	1.0000080217871700	-2.6021676690551300	-0.6660408087618770	-0.009962264691206770	0.6247104314336230	2.560837291726870
bottom_precipitation	62331.0	-7.29568514682043E-18	1.0000080217870700	-2.4987477314825500	-0.7416578651815840	-0.4198599664106990	0.4297353790190580	2.18682524532002
bottom_air_temperature	62331.0	-1.51020682539183E-15	1.0000080217872000	-2.6165763817393700	-0.6057639359408080	0.040473061051873700	0.7347776945915630	2.7455901403901200
bottom_ocean_temperature	62331.0	9.55005185718794E-15	1.0000080217872200	-2.3320956334732000	-0.5781125255816680	-0.09422263753145630	0.5912095463461130	2.345192654237770
lower_right_bed_rock_elevation	62331.0	-1.55033309369934E-17	1.000008021787140	-2.602127672623460	-0.6626741297007140	-0.006833916642421470	0.6302948989144490	2.569748441837190
lower_right_precipitation	62331.0	-5.65415598878583E-17	1.0000080217872500	-2.4944369819078600	-0.7361598346623910	-0.4201849998285460	0.4360249301679190	2.1943020774133800
lower_right_air_temperature	62331.0	-4.88810904836969E-16	1.0000080217872100	-2.502341428591040	-0.5561356297743020	0.05500843837134230	0.741334902770191	2.3722822622593700
lower_right_ocean_temperature	62331.0	6.56465749510902E-14	1.0000080217871800	-2.2449609704576200	-0.5408241743883990	-0.07260436728837180	0.5952670229910180	2.2994038190602000
bed_rock_elevation_3_years_ago	62331.0	1.3223429328612E-17	1.0000080217871700	-2.5823734119320700	-0.6558329831619930	-0.004745535166188570	0.628527302684723	2.5550677314548000
precipitation_3_years_ago	62331.0	-3.64784257341021E-18	1.0000080217872300	-2.522424548138100	-0.7488994919322910	-0.4291017335558970	0.433450545538249	2.2069756017440600
air_temperature_3_years_ago	62331.0	-2.26531023808774E-15	1.0000080217871800	-2.724119610699840	-0.65280622222167	0.04986512430365110	0.7280693700971070	2.7869360102044900
ocean_temperature_3_years_ago	62331.0	3.03281631553325E-14	1.0000080217871900	-2.405951951336530	-0.6107811037733710	-0.11315851290870600	0.58599946126871	2.3811703088317900
bed_rock_elevation_2_years_ago	62331.0	1.82392128670511E-18	1.0000080217871900	-2.5950375800454400	-0.6605617566565560	-0.004596514653163650	0.6290887922693660	2.5635646156582500
precipitation_2_years_ago	62331.0	-1.51385466796524E-16	1.0000080217872300	-2.5328917757462400	-0.7523007231742730	-0.43119875842186500	0.43475997854037000	2.2153510311123300
air_temperature_2_years_ago	62331.0	1.05057866114214E-15	1.0000080217871700	-2.8839874039041800	-0.7125507228138930	0.034745209842898700	0.7350737312463010	2.905868285217880
ocean_temperature_2_years_ago	62331.0	-5.26675510748967E-14	1.000008021787210	-2.5214430749738900	-0.656820506889562	-0.14072932443144500	0.5862612051666270	2.4508837732508700
bed_rock_elevation_1_year_ago	62331.0	1.27674490069358E-17	1.0000080217871700	-2.592970547093830	-0.6598429101527830	-0.006439273363259820	0.628908847807915	2.5620364847489600
precipitation_1_year_ago	62331.0	-2.37109767271664E-17	1.000008021787120	-2.5433372351308100	-0.7567283935567540	-0.43174123765272600	0.4343441674926200	2.2209530090666800
air_temperature_1_year_ago	62331.0	1.31322332642768E-16	1.000008021787180	-3.065001645315100	-0.7787040319839740	0.019247880847550300	0.7454943769034390	3.031791990234560
ocean_temperature_1_year_ago	62331.0	-2.61185528256171E-14	1.000008021787180	-2.6493371272234200	-0.7083617823230990	-0.17093850307052000	0.5856217809438140	2.5265971258442300
ice_velocity	62331.0	92.60670759125770	367.6631323825370	0.0	2.62156379223	8.29279899597	28.34227180485	19250.7460938

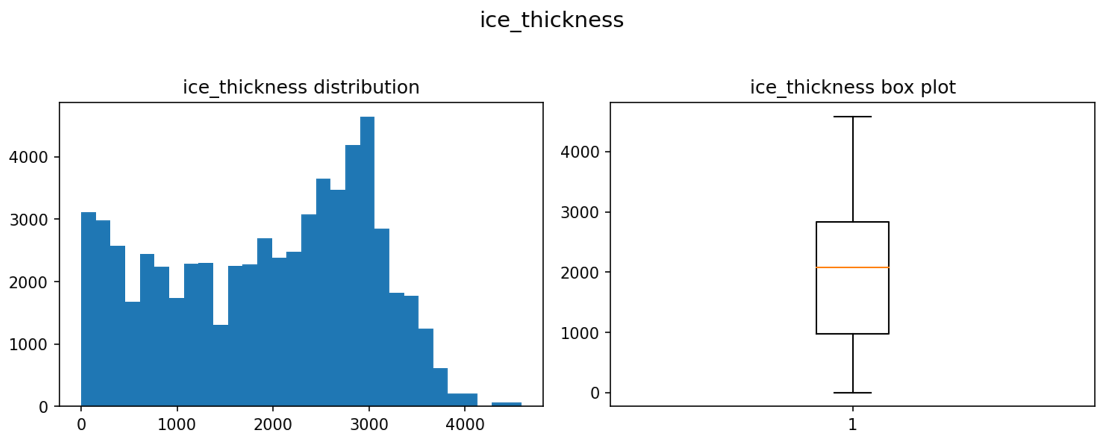
5.4.4 Preprocessed Low-res-more-variables dataset with ice_velocity as the label

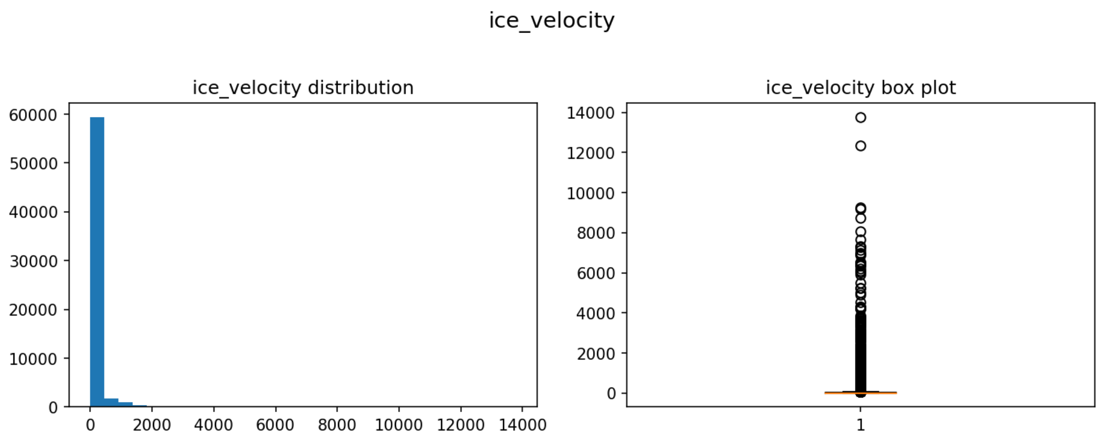
Below are the preprocessed distribution plots of several variables from the Low-res-less-variables dataset and Low-res-more-variables dataset:


air_temperature
air_temperature distribution
air_temperature box plot

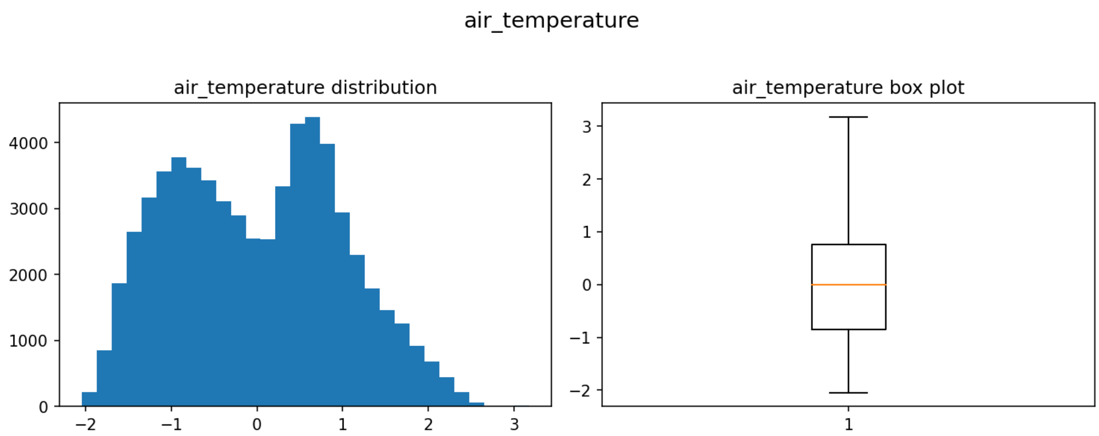

precipitation
precipitation distribution
precipitation box plot

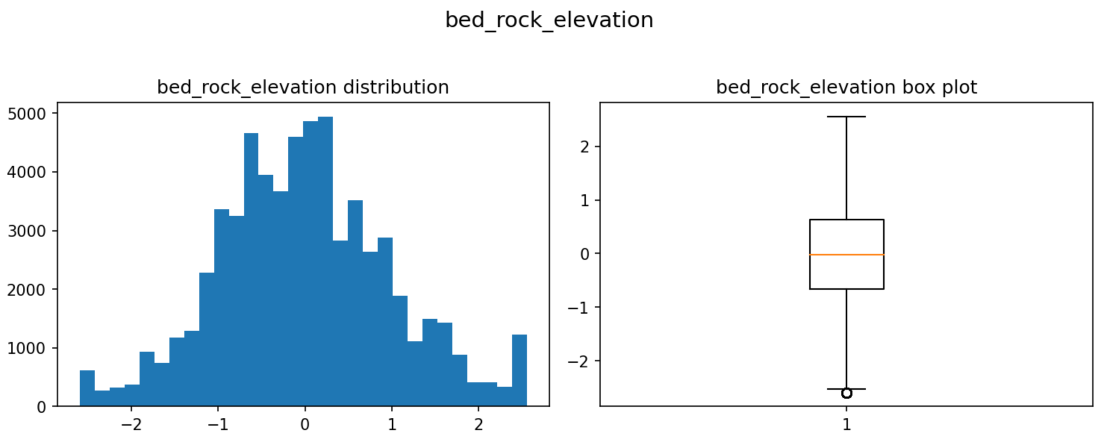

ocean_temperature
ocean_temperature distribution
ocean_temperature box plot

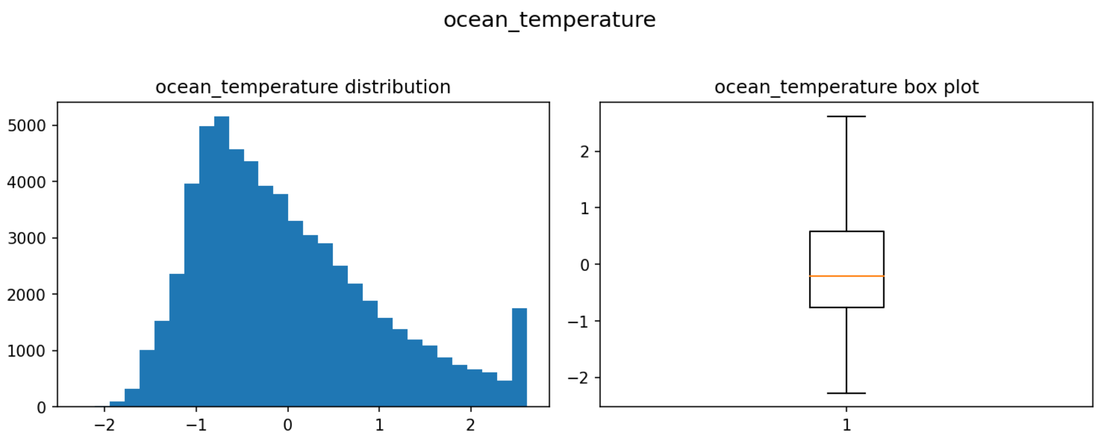


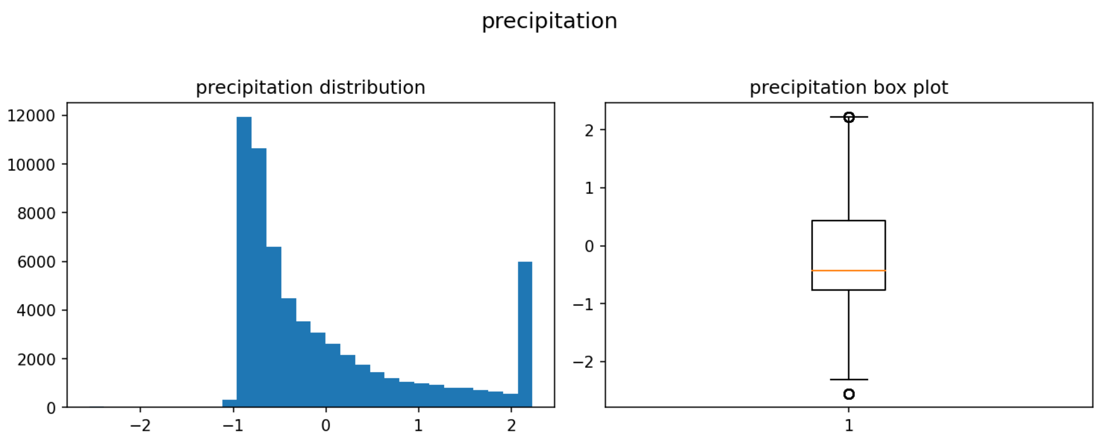


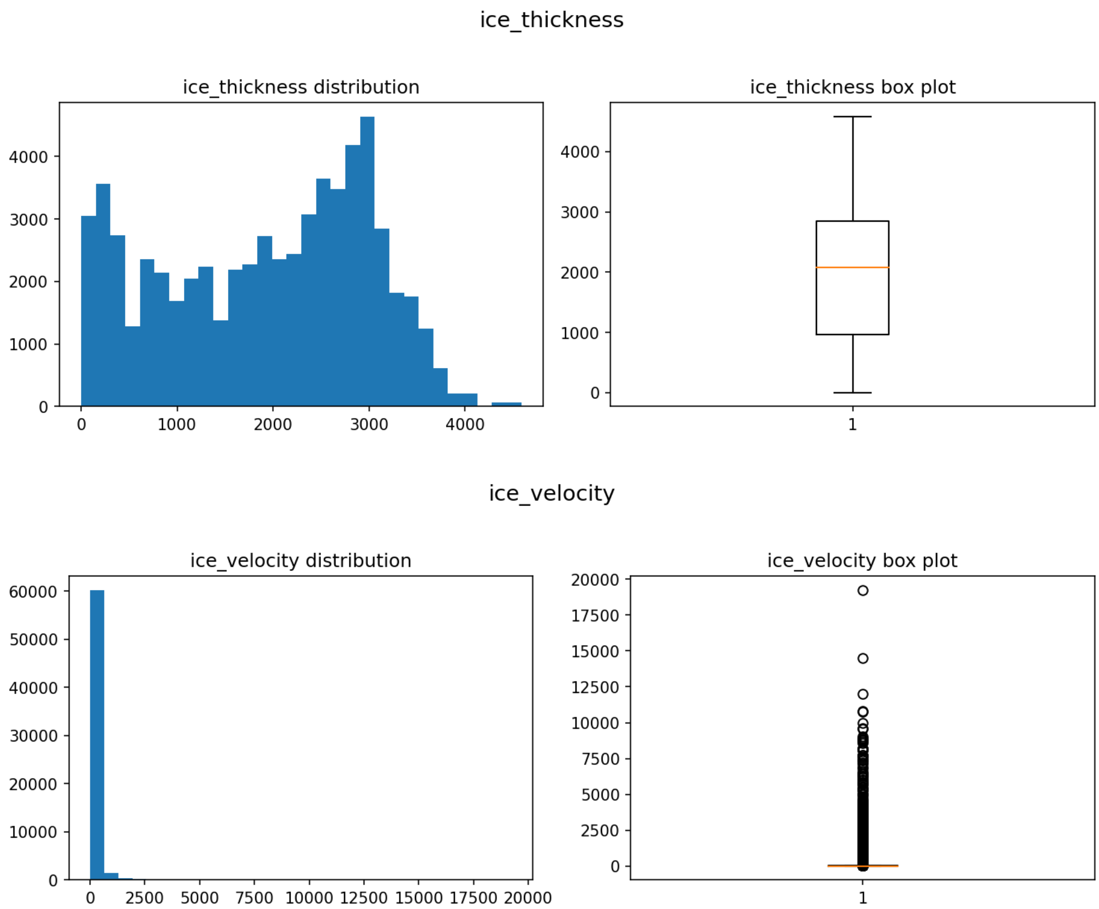
5.4.5 Preprocessed distribution plots from the Low-res-less-variables dataset




bed_rock_elevation
bed_rock_elevation distribution
bed_rock_elevation box plot
5000
4000
3000
2000
1000
0
−2
−1
0
1
2
1

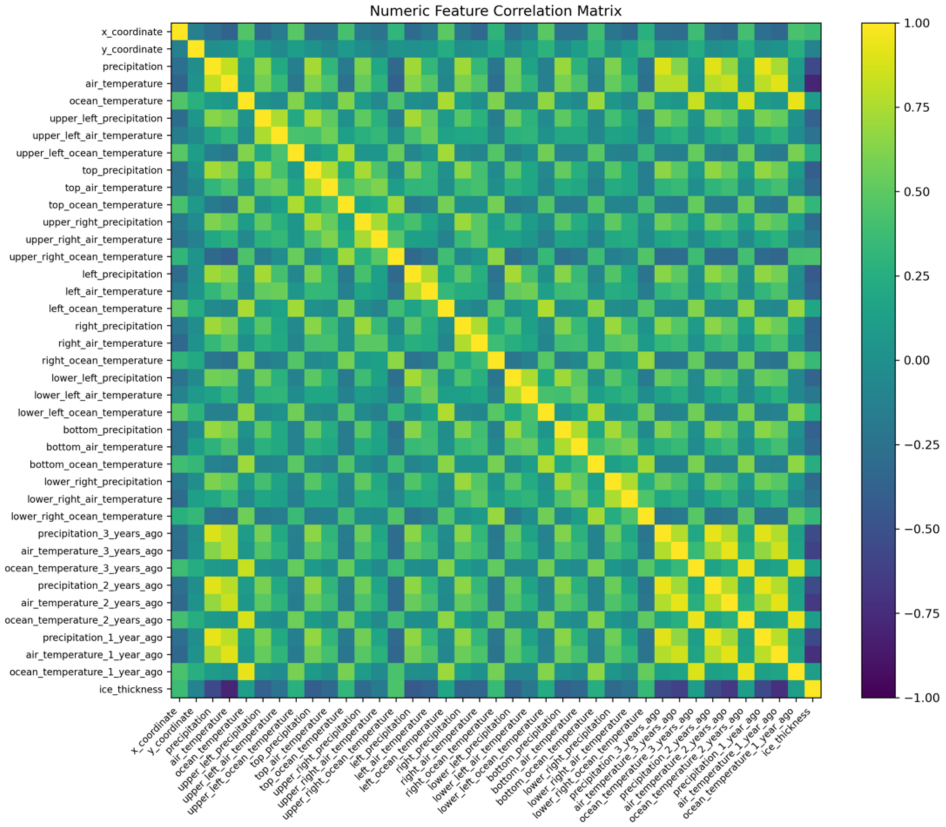

ocean_temperature
ocean_temperature distribution
ocean_temperature box plot
5000
4000
3000
2000
1000
0
−2
−1
0
1
2
1


precipitation
precipitation distribution
precipitation box plot
12000
10000
8000
6000
4000
2000
0
−2
−1
0
1
2
1

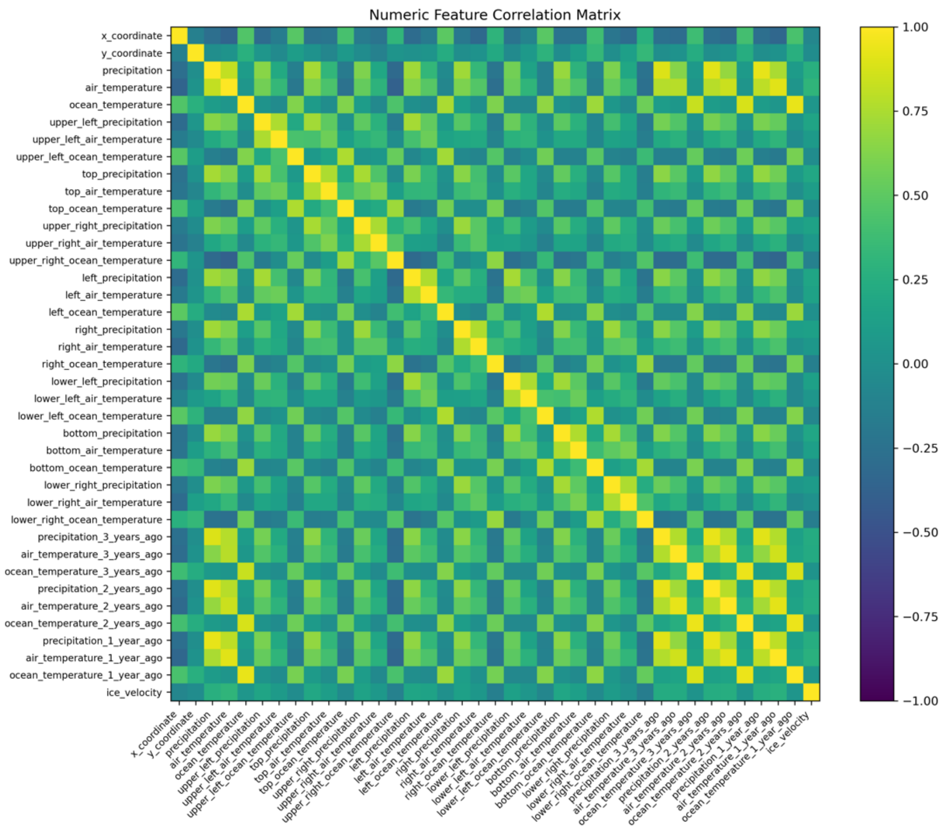


5.4.6 Preprocessed distribution plots from the Low-res-more-variables dataset

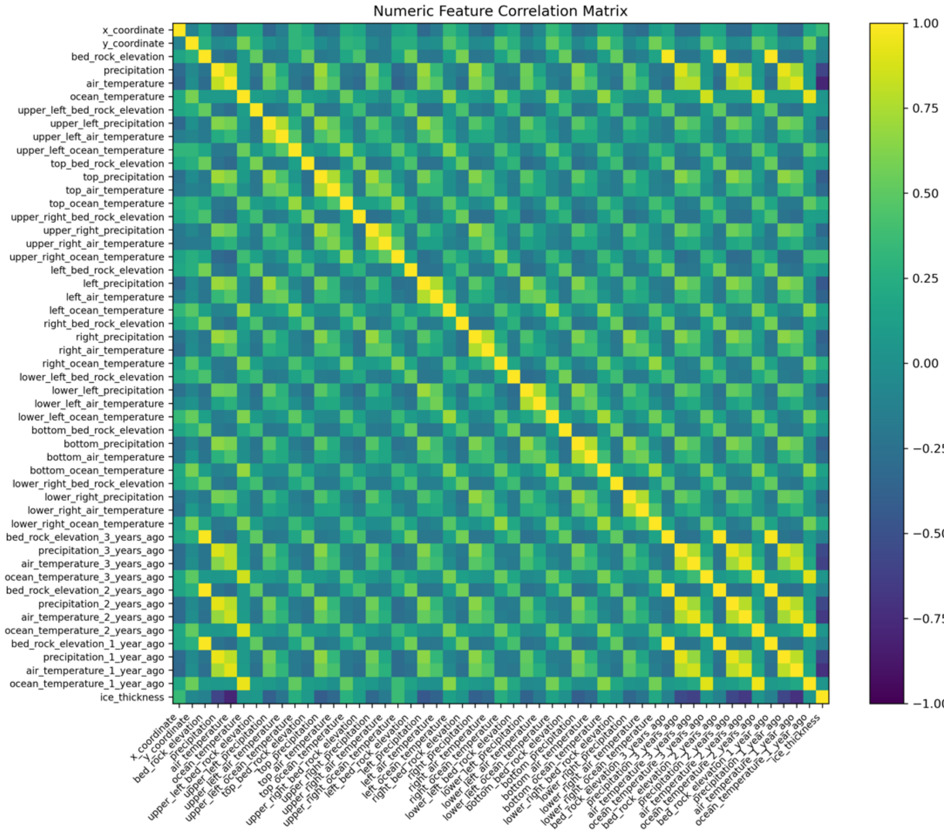
Below are the results of correlation heat map of the preprocessed data from the Low-res-less-variables and Low-res-more-variables datasets using ice_thickness and ice_velocity as the labels, respectively:



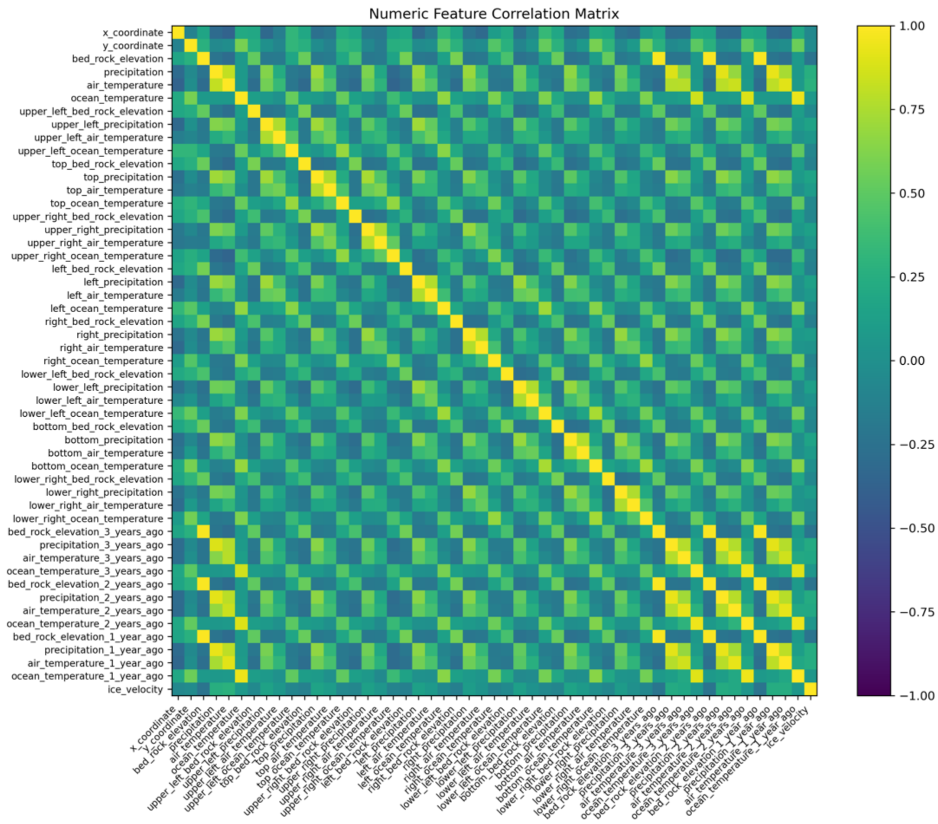
5.4.7 Correlation heat map of preprocessed Low-res-less-variables dataset with ice_thickness



5.4.8 Correlation heat map of preprocessed Low-res-less-variables dataset with ice_velocity



5.4.9 Correlation heat map of preprocessed Low-res-more-variables dataset with ice_thickness



5.4.10 Correlation heat map of preprocessed Low-res-more-variables dataset with ice_ velocity

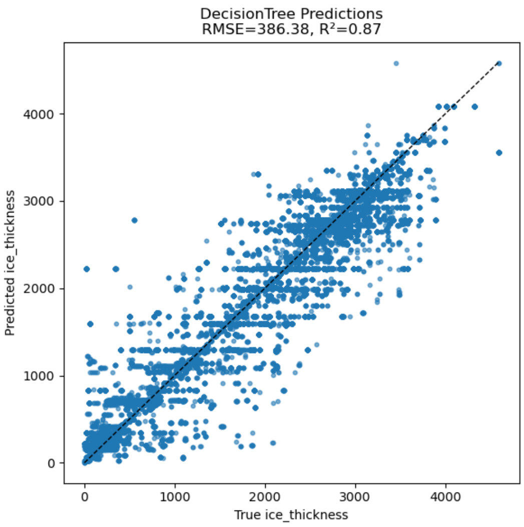
6. MODEL TRAINING AND EVALUATION

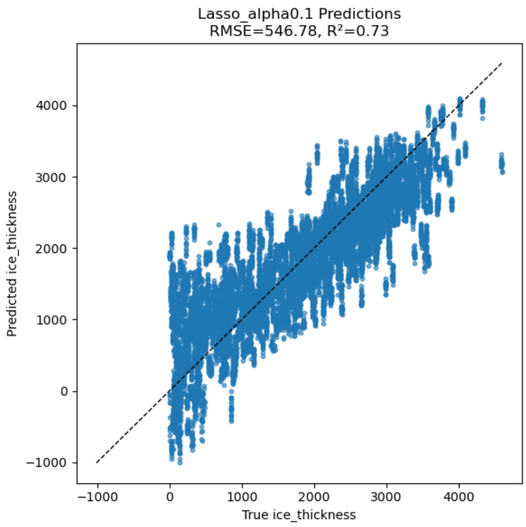
Used the following six regression models to train on the previously engineered feature set, generate predictions, and evaluate their performance:

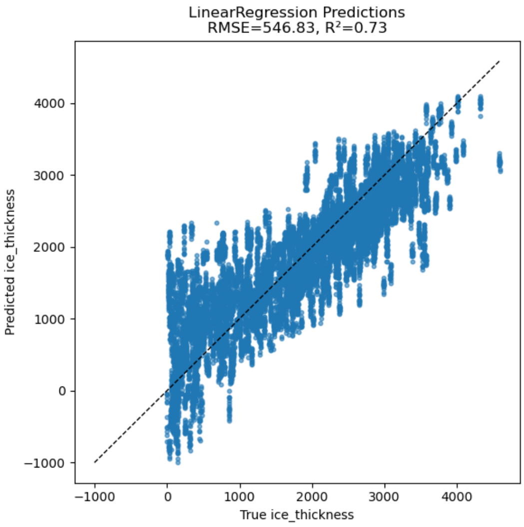
Model Name	Python Class	Key Hyperparameters
LinearRegression	sklearn.linear_model.LinearRegression	(default settings, no regularization)
Lasso_alpha0.1	sklearn.linear_model.Lasso	alpha=0.1, max_iter=10000
SVR_RBF	sklearn.svm.SVR	kernel='rbf', C=1.0, epsilon=0.1
DecisionTree	sklearn.tree.DecisionTreeRegressor	max_depth=10, random_state=<seed>
RandomForest	sklearn.ensemble.RandomForestRegressor	n_estimators=100, max_depth=10, random_state=<seed>, n_jobs=-1
MLPRegressor	sklearn.neural_network.MLPRegressor	hidden_layer_sizes=(100,50), activation='relu', solver='adam', max_iter=500, random_state=<seed>

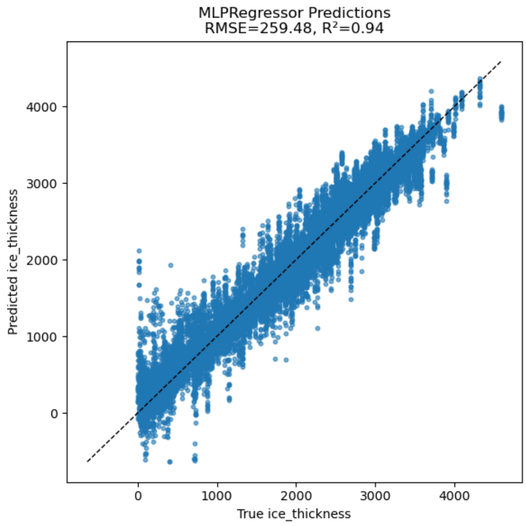
A fixed random seed of 42 was used; the model was trained on the entire training set and evaluated on the entire test set. The evaluation metrics used are Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). Lower RMSE values denote better performance, and R^2 values closer to 1 indicate a superior fit.

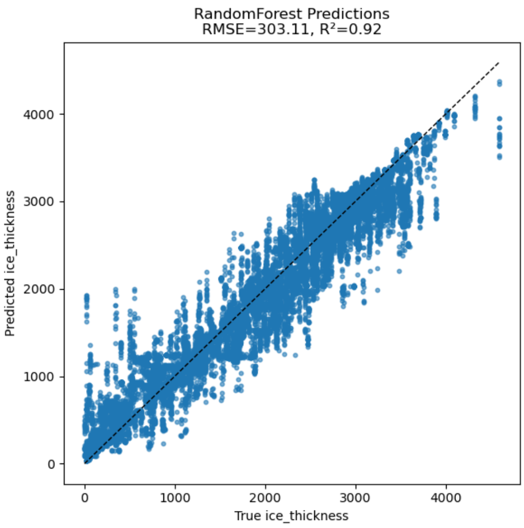
Below are the prediction results and performance metrics of each model on the Low-Res Less-Variables dataset, using ice_thickness as the target label:

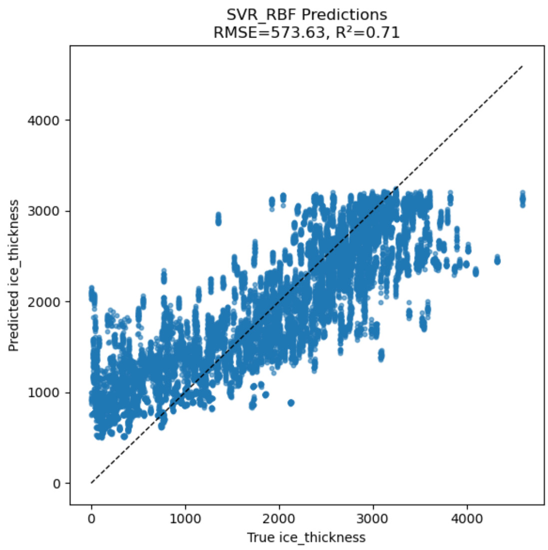






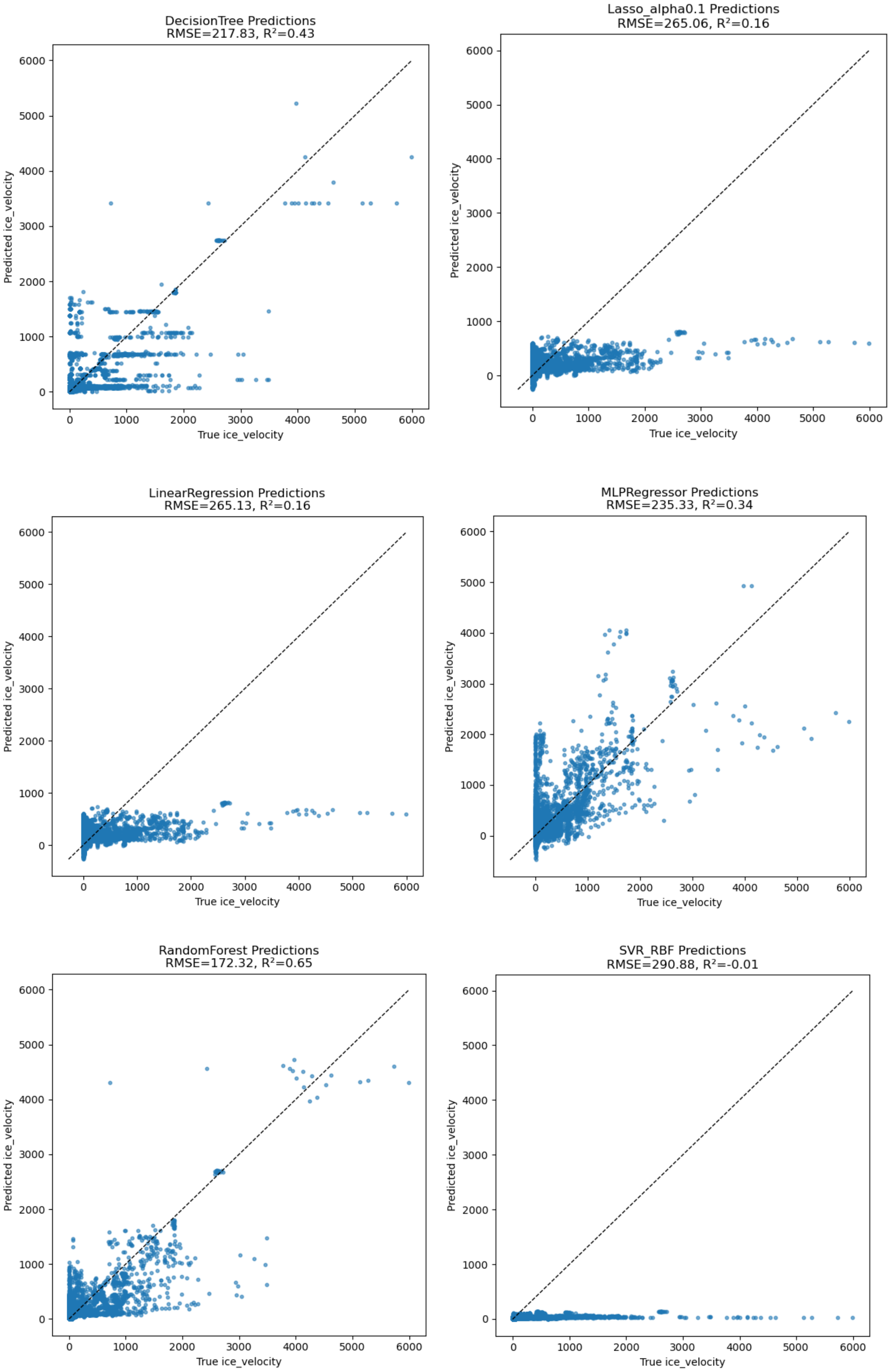






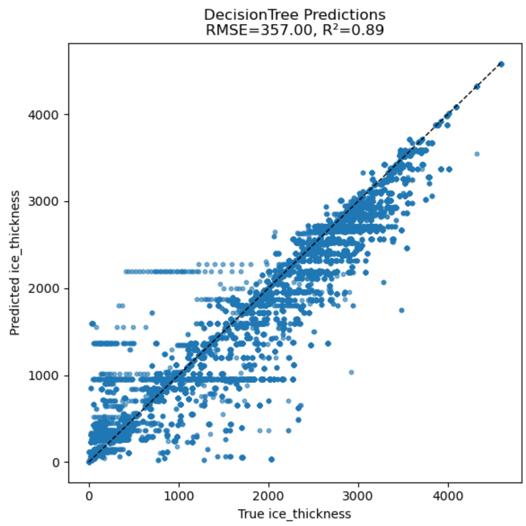
Model Name	**RMSE**	**R^2**
LinearRegression	546.830	0.733
Lasso_alpha0.1	546.780	0.733
SVR_RBF	573.634	0.706
DecisionTree	386.384	0.867
RandomForest	303.113	0.918
MLPRegressor	259.484	0.940

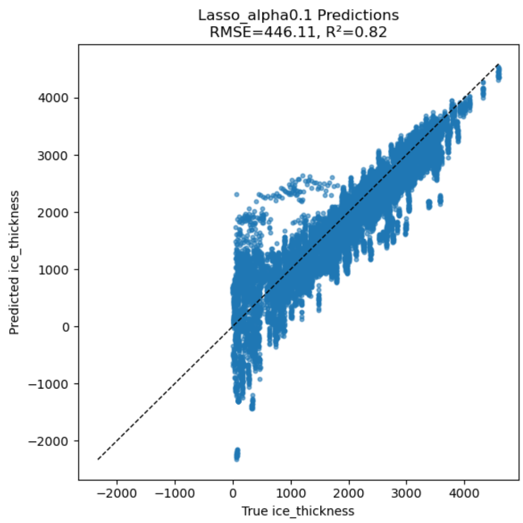
Below are the prediction results and performance metrics of each model on the Low-Res Less-Variables dataset, using ice_velocity as the target label:

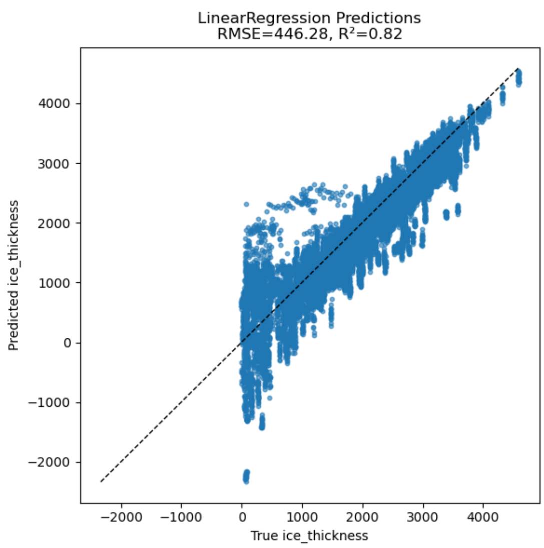

DecisionTree Predictions
RMSE=217.83, R²=0.43
Lasso_alpha0.1 Predictions
RMSE=265.06, R²=0.16
LinearRegression Predictions
RMSE=265.13, R²=0.16
MLPRegressor Predictions
RMSE=235.33, R²=0.34
RandomForest Predictions
RMSE=172.32, R²=0.65
SVR_RBF Predictions
RMSE=290.88, R²=-0.01
Predicted ice_velocity
True ice_velocity

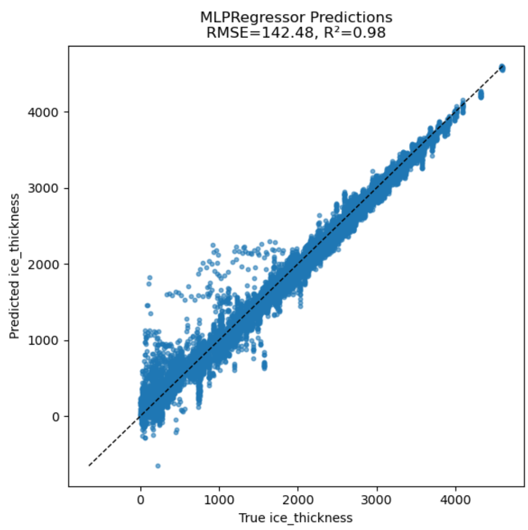
Model Name	RMSE	R^2
LinearRegression	265.132	0.163
Lasso_alpha0.1	265.059	0.163
SVR_RBF	290.879	-0.008
DecisionTree	217.832	0.435
RandomForest	172.320	0.646
MLPRegressor	235.332	0.341

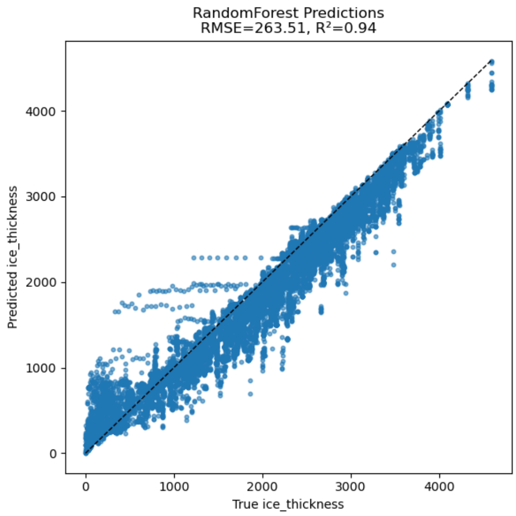
Below are the prediction results and performance metrics of each model on the Low-Res-More-Variables dataset, using ice_thickness as the target label:

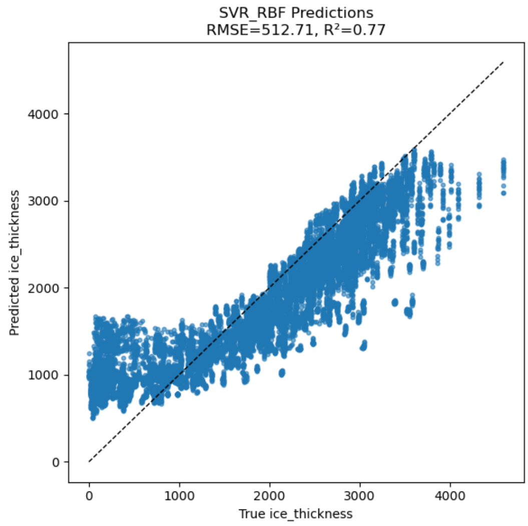






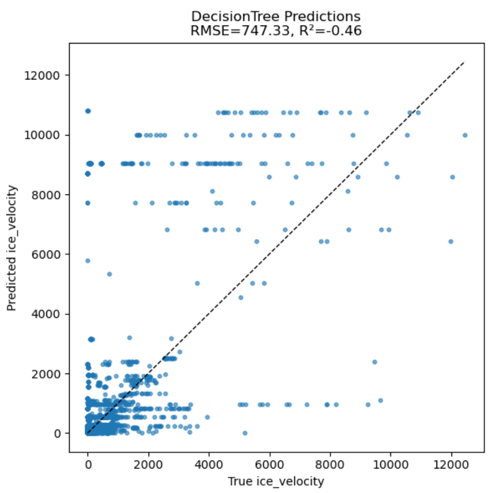


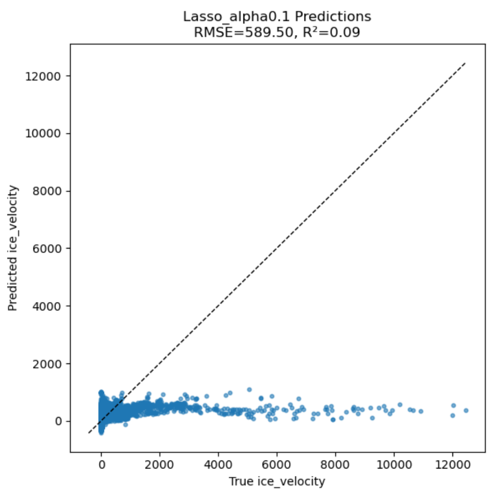


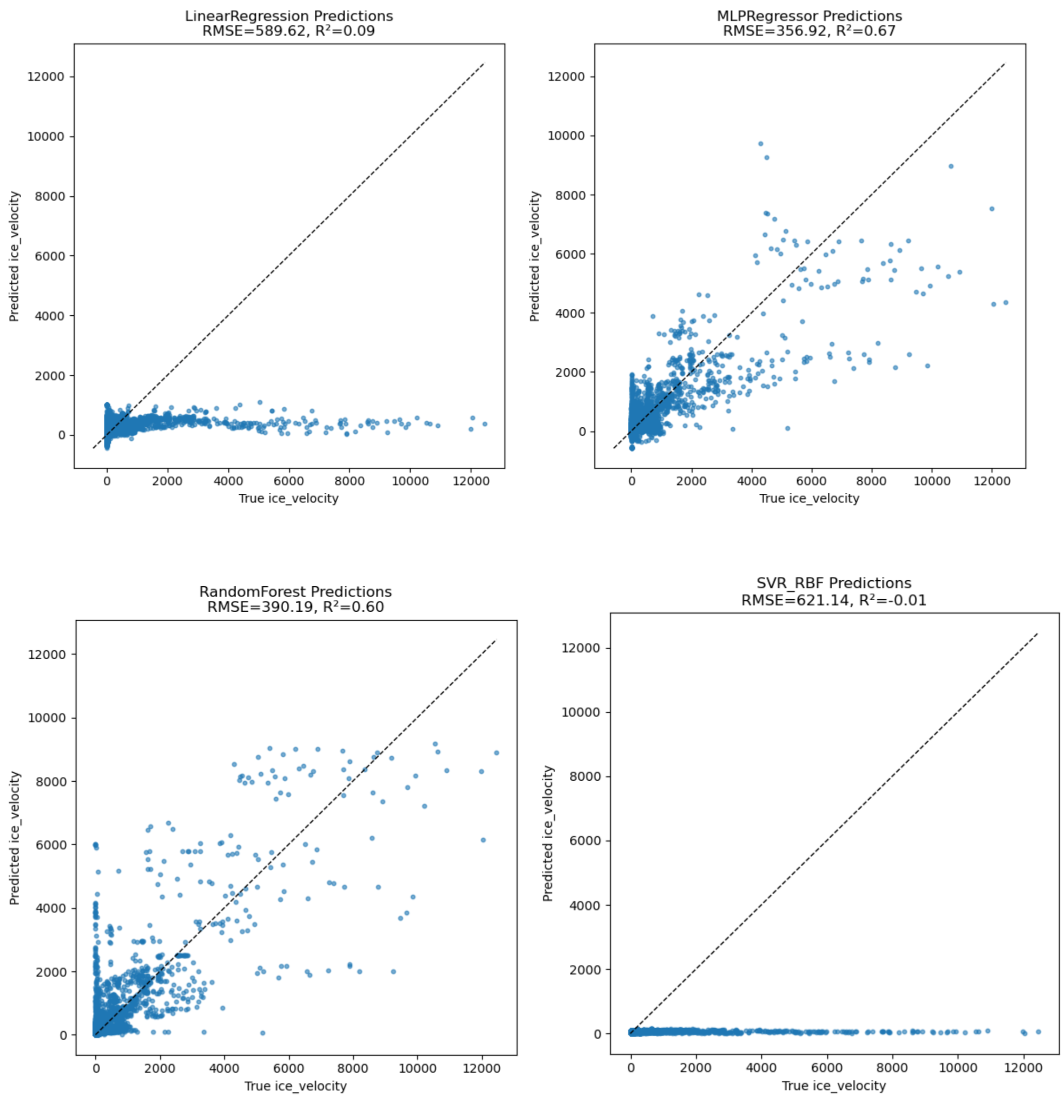


Model Name	RMSE	R^2
LinearRegression	446.279	0.823
Lasso_alpha0.1	446.109	0.823
SVR_RBF	512.705	0.766
DecisionTree	357.001	0.886
RandomForest	263.508	0.938
MLPRegressor	142.475	0.982

Below are the prediction results and performance metrics of each model on the Low-Res-More-Variables dataset, using ice_velocity as the target label:







Model Name	RMSE	R^2
LinearRegression	589.621	0.093
Lasso_alpha0.1	589.498	0.093
SVR_RBF	621.137	-0.007
DecisionTree	747.326	-0.458
RandomForest	390.185	0.603
MLPRegressor	356.918	0.667

The figure above compares each model's performance in terms of RMSE and R^2. Overall, nonlinear models (Decision Tree, Random Forest, MLP) clearly outperform linear models (LinearRegression, Lasso). Moreover, in most cases, MLPRegressor achieves the best results for both

RMSE and R^2, with Random Forest a close second. This indicates that the data contains inherent nonlinear relationships, which is why nonlinear models generally outperform linear ones.

7. USING GENETIC ALGORITHM (GA) TO OPTIMIZE FEATURE SELECTION

According to the previous correlation heat map, many features exhibit strong correlations, which can introduce redundancy during training. Used a GA to optimize feature selection.

The GA is designed as follows:

- Representation:

Construct a binary vector of length n (a 0/1 array) to indicate whether each feature is selected, where n is the total number of candidate features. Specifically:

- Each chromosome (individual) is a binary array of the form $[b_0, b_1, \ldots, b_{n-1}]$.
- If the i-th bit $b_i = 1$, it means feature i is selected; if $b_i = 0$, that feature is skipped.

For example, if there are 12 candidate features in total, one possible chromosome could be:

[1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 1]

This indicates that features 0, 2, 4, 5, 8, and 11 (6 out of 12) are selected.

- Fitness Function:

The fitness function is the root mean squared error (RMSE) on the validation set. For each chromosome (feature subset), the corresponding columns are extracted from X_train and used to train an MLPRegressor. The model then makes predictions on X_val and the RMSE of those predictions is computed. The GA's objective is to find the chromosome that minimizes this validation RMSE.

- Crossover, Mutation, and Selection Operators:

Due to the limited computing power of my laptop, the population size is set to pop_size = 20 and the number of generations is set to N_GEN = 10.

- **Initialization (Sampling):**
  Use BinaryRandomSampling() to generate a random binary population matrix of shape (pop_size × n_features).
- **Crossover:**
  Apply the two-point crossover operator TwoPointCrossover(prob=CROSSOVER_PROB). prob is set to 0.8 indicates that each pair of parents has an 80% chance of undergoing two-point crossover; otherwise, the parents are copied unchanged.
- **Mutation:**
  Apply the bit-flip mutation operator BitflipMutation(prob=MUTATION_PROB). prob is set to 0.1 indicates that each gene (bit) has a 10% chance of being flipped (0 → 1 or 1 → 0).
- **Selection and Replacement:**
  Use the default tournament selection combined with elitism to form the next generation.

The GA optimization is implemented in the GAFeatureSelection.py script. The previously generated preprocessed_train and preprocessed_test datasets are loaded. The preprocessed_train data is split into training and validation sets according to the VALID_SIZE ratio for GA training. The model and parameters used are the same as in earlier experiments: MLPRegressor(hidden_layer_sizes=(100,50), activation='relu', solver='adam', max_iter=500, early_stopping=True, random_state=RANDOM_SEED). Once training is complete, the selected feature subset is printed. That subset is then applied to both the full training set and the full test set. Finally, an MLPRegressor is trained on the entire training set (using only the chosen features) and evaluated on the entire test set.

Below are the results on the Low-Res Less-Variables dataset using ice_thickness as the label:

```
=================================================
n_gen  |  n_eval  |     f_avg     |     f_min
=================================================
     1 |       20 |  2.804704E+02 |  2.353550E+02
     2 |       40 |  2.518273E+02 |  2.353550E+02
     3 |       60 |  2.402275E+02 |  2.188906E+02
     4 |       80 |  2.302011E+02 |  2.177304E+02
     5 |      100 |  2.247263E+02 |  2.109440E+02
     6 |      120 |  2.202056E+02 |  2.041690E+02
     7 |      140 |  2.171998E+02 |  2.041690E+02
     8 |      160 |  2.141827E+02 |  2.041690E+02
     9 |      180 |  2.127665E+02 |  2.041690E+02
    10 |      200 |  2.104123E+02 |  1.990949E+02

=== GA Feature Selection Results ===
Best RMSE (validation set): 199.0949
Number of selected features: 27 / 38
List of selected features:
  - x_coordinate
  - y_coordinate
  - precipitation
  - air_temperature
  - upper_left_precipitation
  - upper_left_air_temperature
  - top_precipitation
  - upper_right_air_temperature
  - upper_right_ocean_temperature
  - left_precipitation
  - left_air_temperature
  - left_ocean_temperature
  - right_air_temperature
  - right_ocean_temperature
  - lower_left_air_temperature
  - lower_left_ocean_temperature
  - bottom_precipitation
  - bottom_air_temperature
  - lower_right_precipitation
  - lower_right_air_temperature
  - lower_right_ocean_temperature
  - precipitation_3_years_ago
  - air_temperature_3_years_ago
  - precipitation_2_years_ago
  - air_temperature_2_years_ago
  - precipitation_1_year_ago
  - ocean_temperature_1_year_ago

Test Set RMSE: 256.8674

Test Set r2: 0.9411
```

Based on the experimental results, the new feature combination yields improved performance on both RMSE and R^2 metrics (RMSE is now 256.8674 versus 259.484 previously, and R^2 is now 0.9411 versus 0.940 previously). However, because both pop_size and N_GEN were set relatively low, the results indicate that the model has not yet converged. These findings serve as a proof-of-concept implementation of one possible GA-based approach.

- Future Work:

Parameter Tuning: Key GA hyperparameters (pop_size, N_GEN, MUTATION_PROB, and CROSSOVER_PROB) can be further optimized. Adjusting these values (for example, increasing population size and N_GEN) should improve convergence and overall performance.

Model and its hyperparameter tuning: Further refinement of the model architecture and optimization of the hyperparameters can be done to obtain better prediction performance.

Initialization Enhancement: The current initial population is generated randomly via BinaryRandomSampling(). It would be beneficial to incorporate domain knowledge by preselecting certain correlated features into the initial population. By seeding GA with these informed individuals, the search process may converge more quickly and discover higher-quality feature subsets.

Shuffling the Training Set: Since the train/test split was performed chronologically by year, it would be advantageous to randomize the training data before fitting the model in order to eliminate any ordering bias. Although MLPRegressor enables shuffling by default at each epoch, other algorithms may require explicit configuration.